

Disorders of Calcium, Magnesium, and Phosphate Balance

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DISORDERS OF CALCIUM HOMEOSTASIS

The extracellular fluid (ECF) calcium concentration in the human body is tightly regulated by a complex process. Three organs—skeleton, kidney, and intestine—are involved in this process through their direct or indirect interaction with parathyroid hormone (PTH), parathyroid hormone-related peptide (PTHrP), vitamin D, and calcitonin. Phosphatonins such as fibroblast growth factor 23 (FGF-23), although they participate in phosphate and vitamin D homeostasis, do not directly modify extracellular calcium.

This homeostatic system is modulated by dietary and environmental factors, including vitamins, hormones, medications, and mobility. Disorders of extracellular calcium homeostasis may be regarded as perturbations of this homeostatic system, either at the level of the genes controlling this system (e.g., as in familial hypocalciuric hypercalcemia, pseudohypoparathyroidism, or vitamin D-dependent rickets) or perturbations of this system induced by nongenetic means (e.g., as in lithium toxicity or postsurgical hypoparathyroidism).

Calcium fluxes between the ECF and one of the organs (skeleton, kidney, intestine) or their combination, as well as abnormal binding of the calcium to serum protein, can cause hypercalcemia. PTH directly protects against hypocalcemia by augmenting calcium mobilization from bone, increasing kidney tubular reabsorption of calcium, and enhancing intestinal absorption of calcium. PTH also indirectly protects against hypocalcemia through its effect on vitamin D metabolism. The phosphate mobilization from bone that is induced by PTH is countered by PTH-mediated decreases in kidney tubular reabsorption of phosphate. States with excess PTH may cause hypercalcemia, whereas PTH deficiency is associated with hypocalcemia. Similarly, PTHrP promotes bone resorp-

tion, enhances kidney reabsorption of calcium, and decreases kidney tubular reabsorption of phosphate. Excess of this hormone is responsible for the hypercalcemia of malignancy. Vitamin D and its metabolites increase intestinal absorption of calcium and cause bone resorption; therefore, excess vitamin D would induce hypercalcemia. Calcitonin inhibits bone resorption, but its physiologic role in the protection against hypercalcemia in humans is not proven.

WHOLE-BODY CALCIUM HOMEOSTASIS

An adult human body contains approximately 1000 to 1300 g of calcium, with 99.3% in bone and teeth as hydroxyapatite crystal, 0.6% in soft tissues, and 0.1% in ECF, including 0.03% in plasma.¹ Intracellular concentrations of free calcium are very low (~100 nM) compared with extracellular concentrations of free calcium (~1 mM). Indeed, a 10,000-fold gradient is present. Maintenance of normal calcium balance and serum calcium levels depends on the integrated regulation of calcium absorption and secretion by the intestinal tract, excretion of calcium by the kidney, and calcium release from and deposition into bone. In young adults, calcium balance is neutral. Approximately 1000 mg of calcium is ingested per day, 200 mg absorbed by the gut, mainly the duodenum, and 800 mg excreted via the gut. Of 10 g of calcium filtered by the kidney daily, only approximately 200 mg is excreted in the urine. At the same time, 500 mg of calcium is released from bone, and the same amount is deposited with new bone formation. PTH, by stimulating bone resorption and distal tubular calcium reabsorption in the kidney and by activating kidney hydroxylation of 25(OH)D₃ to 1,25(OH)₂D₃, increases serum calcium levels. Depression in serum levels of calcium, by itself, stimulates, through the calcium-sensing receptor (CaSR) in the parathyroid gland, the secretion of preformed

PTH from the parathyroid gland within seconds. Subsequently, PTH biosynthesis by the parathyroid gland increases over 24 to 48 hours and, if sustained chronically, is followed by parathyroid gland hypertrophy and hyperplasia. Vitamin D metabolites, serum phosphorus, and FGF-23 levels also regulate PTH levels in blood.

The values for total serum calcium concentration in adults vary among clinical laboratories, depending on the methods of measurement, with the normal range being between 8.6 and 10.3 mg/dL (2.15–2.57 mmol/L).^{2,3} Variations in serum calcium levels occur, depending on age and gender, with a general trend for a lower serum calcium level with aging.⁴

Calcium in blood exists in three distinct fractions—protein-bound calcium (40%), free (ionized) calcium (48%), and calcium complexed to various anions, such as phosphate, lactate, citrate, and bicarbonate (12%).⁵ The latter two forms, complexed calcium and free calcium ion, together comprise the fraction of plasma calcium that can be filtered. Plasma albumin is responsible for 90% and globulins for 10% of protein-bound calcium. Free calcium is the physiologically active component of extracellular calcium with regard to CaSR signaling, cardiac myocyte contractility, neuromuscular activity, bone mineralization, and other calcium-dependent processes. It is measured in most hospitals using ion-selective electrodes; values in adults range from 4.65 to 5.28 mg/dL (1.16–1.32 mmol/L).^{4,6} Total calcium reflects the levels of free calcium if plasma levels of protein, pH, and anions are normal.

The relationship between calcium ion and the concentration of protein in the serum is represented by a simple mass action expression:

$$([\text{Ionized Ca}^{2+}] \times [\text{protein}]) / (\text{calcium proteinate}) = K$$

where [protein] equals the concentration of serum proteins, primarily albumin. Because K is a constant, the numerator and denominator must change proportionately in any physiologic or pathologic state. A change in the concentration of total serum calcium will occur after a change in the concentration of serum proteins or alterations in their binding properties and after a primary change in the concentration of calcium ion. A fall in the serum albumin level reduces the protein and calcium proteinate levels proportionately, resulting in a fall in the total serum calcium level, with the free calcium ion concentration remaining normal. If plasma levels of albumin are low, an adjustment of the measured serum levels of calcium should be made (commonly but erroneously referred to as a “correction”). For the routine clinical interpretation of serum calcium needed for appropriate care of patients, a simple formula for adjustment of the total serum calcium concentration for changes in plasma albumin concentration is used by clinicians.

In conventional units:

$$\text{Adjusted total calcium (mg/dL)} = \text{total calcium (mg/dL)} + 0.8(4 - \text{albumin [g/dL]})$$

In SI units:

$$\text{Adjusted total calcium (mmol/L)} = \text{total calcium (mmol/L)} + 0.002(40 - \text{albumin [g/dL]})$$

This formula was endorsed in 1977 by an editorial in the *British Medical Journal*⁷; the correction factor of 0.02 (in SI units) was chosen arbitrarily for simplicity from the range

available in the literature at that time (0.018–0.025). This adjustment can also correct for errors in measurement of total calcium related to the hemoconcentration of a blood sample because of the prolonged use of a tourniquet or because of hemodilution when blood is drawn in hospitalized patients in a supine position.³ Other formulas have been developed, particularly for chronic kidney disease (CKD) patients, that have a slightly better (although not statistically significant) discriminatory ability to make the diagnosis of hypocalcemia or hypercalcemia, as established from the measurement of free calcium.^{8,9} Also, a fall in pH of 0.1 unit will cause approximately a 0.1-mEq/L rise in the concentration of ionized calcium, because hydrogen ion displaces calcium from albumin, whereas alkalosis decreases free calcium by enhancing the binding of calcium to albumin.⁴ There is no correction for this effect of pH in the previous formula, which also limits its accuracy.

Calcium binding to globulin is low (1.0 g of globulin binds 0.2–0.3 mg of calcium), and it is unusual to see a change in the total concentration of serum calcium as a result of alterations in the globulin levels in blood. However, in cases in which the globulin concentration in serum is extremely high (>8.0 g/dL), such as in multiple myeloma, a mild to moderate hypercalcemia may be seen because of an elevation of the globulin-bound calcium. In addition, immunoglobulin G (IgG) myeloma proteins may have increased calcium-binding properties, and an elevation in the total level of serum calcium could occur, even with a moderate increase in serum levels of globulins. In these cases, the ionized calcium in serum is normal; therefore, this type of hypercalcemia would not require treatment.

Unfortunately, calcium status will be incorrectly predicted by this formula in 20% to 30% of subjects,¹⁰ and the agreement between corrected and free calcium is only fair.¹¹ Thus, free calcium should be assessed, particularly in critically ill patients with acid-base disturbances, in patients exposed to large amounts of citrated blood, and in those with severe blood protein disorders. Patients with CKD and those treated with dialysis may also benefit from free calcium measurements in an evaluation of their mineral bone metabolism status.¹² It is important to recognize that free calcium results in blood are affected by factors related to the handling of specimens, including duration of cellular metabolism, loss of CO₂, and use of anticoagulants.²

HYPERCALCEMIA

Hypercalcemia is relatively common and frequently overlooked, with an annual incidence estimated to be about 0.1% to 0.2% and a prevalence of 0.17% to 2.92% in a hospital population and 1.07% to 3.9% in a normal population.¹³

Hypercalcemia results from an alteration in the net fluxes of calcium to and from four compartments—bone, gut, kidney, and serum-binding proteins (Box 18.1). Usually, the hypercalcemia is caused by net calcium movement from the skeleton into ECF through increased osteoclastic bone resorption, as in hyperparathyroidism (HPT) or excess PTHrP production in malignancy. PTH acts via the PTH receptor 1 on the osteoblasts. PTH receptor 1 is encoded by the *PTH1R* gene. It activates cyclic adenosine monophosphate (cAMP) signaling in osteoblasts and upregulates the expression of receptor activator for nuclear factor-kappa B (RANK) ligand (RANKL)

Box 18.1 Causes of Hypercalcemia

Malignancy-associated hypercalcemia

- Humoral hypercalcemia of malignancy (HHM) with secretion of parathyroid hormone (PTH)-related protein by the tumor
- Local osteolytic hypercalcemia (LOH)
- Tumor (lymphoma, germinoma) generation of $1,25(\text{OH})_2\text{D}$
- Ectopic PTH secretion from tumor

Primary hyperparathyroidism

- Adenoma, hyperplasia, carcinoma
- Multiple endocrine neoplasia types 1 and 2a

Familial hypocalciuric hypercalcemia

Neonatal severe hyperparathyroidism

Other endocrine disorders

- Hyperthyroidism
- Acromegaly
- Pheochromocytoma
- Acute adrenal insufficiency

Granulomatous disorders

- Sarcoidosis
- Tuberculosis
- Berylliosis
- Disseminated coccidioidomycosis or candidiasis
- Histoplasmosis
- Leprosy
- Granulomatous lipid pneumonia
- Silicone-induced granuloma
- Eosinophilic granuloma
- Farmer's lung

Vitamin overdoses

- Vitamin D
- Vitamin A

Immobilization

Kidney disease

- Diuretic phase of acute kidney injury, especially resulting from rhabdomyolysis
- Chronic kidney disease
- After kidney transplantation

Medications

- Milk-alkali syndrome
- Thiazide diuretics
- Lithium
- Fosarnet
- Growth hormone
- Recombinant human PTH (1-34; teriparatide)
- Theophylline and aminophylline toxicity
- Estrogen and selective estrogen receptor modulators (SERMs)
- Vasoactive intestinal polypeptide
- Hyperalimentation regimens

Idiopathic hypercalcemia of infancy

Increased serum protein level

- Hemoconcentration
- Hyperglobulinemia due to multiple myeloma

in osteoblasts. RANKL binds to RANK on osteoclasts. RANK activation on osteoclasts through the RANK-RANKL interaction causes recruitment, proliferation, and activation of osteoclasts for bone resorption. Therefore, it is somewhat curious that PTH exerts a direct effect on osteoblasts and an indirect effect on osteoclasts. An increase in the bone resorption rate without an increase in the bone formation rate will cause hypercalcemia.

Myeloma cells may induce multiple osteoclastogenic factors such as RANKL in nonosteoblastic stromal cells or decrease production of osteoprotegerin, a decoy receptor for RANKL. As with PTH, excess, circulating $1,25\text{-dihydroxyvitamin D}$ ($1,25[\text{OH}]_2\text{D}$) from various causes also activates osteoclastic bone resorption indirectly through osteoblasts. Increased intestinal calcium absorption may lead to the development of hypercalcemia, as in vitamin D overdose or milk-alkali syndrome. In general, the kidney does not contribute to hypercalcemia; rather, it defends against the development of hypercalcemia. Typically, hypercalciuria precedes hypercalcemia. Extracellular calcium itself appears to have a calciuric effect on the kidney tubule by its direct action on the CaSR of the thick ascending limb (TAL). Thus, in most hypercalcemic states, kidney calcium handling is subject to competing influences; excess PTH or PTHrP acts on the PTH-PTHrP receptor to promote kidney calcium reabsorption, and excess calcium acts on the calcium receptor to promote calcium excretion.¹⁴

In rare cases, the kidney can actively contribute to the development of hypercalcemia. As opposed to primary HPT and humoral hypercalcemia of malignancy, in which increases in kidney calcium excretion are observed, kidney calcium excretion is not elevated in familial hypocalciuric hypercalcemia because of a defective kidney response to calcium itself. The hypercalcemia associated with thiazide use is also mediated by the kidney; in both thiazide use and its genetic counterpart, Gitelman syndrome, kidney calcium excretion is decreased.

SIGNS AND SYMPTOMS

Hypercalcemia adversely affects the function of almost all organ systems, but in particular the kidney, central nervous system, and cardiovascular system. The clinical manifestations of hypercalcemia relate more to the degree of hypercalcemia and rate of increase than to the underlying cause. Hypercalcemia may be classified based on the level of total serum calcium:¹⁵

- Mild: $[\text{Ca}] = 10.4$ to 11.9 mg/dL
- Moderate: $[\text{Ca}] = 12.0$ to 13.9 mg/dL
- Severe (hypercalcemic crisis): $[\text{Ca}] = 14.0$ to 16 mg/dL

Much higher levels are occasionally observed. Signs and symptoms and complications of hypercalcemia are summarized in [Box 18.2](#).

As many as 10% of patients with elevated levels of serum calcium are detected by a routine screening test of blood chemistry and are considered to have so-called asymptomatic hypercalcemia. However, even very mild hypercalcemia may be of clinical significance in as much as some studies have suggested an increased cardiovascular risk from mild but prolonged calcium level elevations.¹⁶

In symptomatic patients, the spectrum of the clinical presentation is varied and could be nonspecific. Mild hypercalcemia may present with malaise, weakness, minor joint pain, and other vague symptoms. In patients with severe hypercalcemia, the major symptoms are more likely to be nausea, vomiting, constipation, polyuria, and mental disturbances, ranging from headache and lethargy to coma. Recent loss of memory could be prominent and may be a presenting symptom.

Box 18.2 Clinical Features of Hypercalcemia

General

Malaise, tiredness, weakness

Neuropsychiatric

Impaired concentration, loss of memory, headache, drowsiness, lethargy, disorientation, confusion, irritability, depression, paranoia, hallucinations, ataxia, speech defects, visual disturbances, deafness (calcification of eardrum), pruritus, mental retardation (infants), stupor, coma

Neuromuscular

Muscle weakness, hyporeflexia or absent reflexes, hypotonia, myalgia, arthralgia, bone pain, joint effusion, chondrocalcinosis, dwarfism (infants)

Gastrointestinal

Loss of appetite, dry mouth, thirst, polydipsia, nausea, vomiting, constipation, abdominal pain, weight loss, acute pancreatitis (calcifying), peptic ulcer, acute gastric dilation

Kidney

Polyuria, nocturia, nephrocalcinosis, nephrolithiasis, interstitial nephritis, acute kidney injury and chronic kidney disease

Cardiovascular

Arrhythmia, bradycardia, first-degree heart block, short Q-T interval, bundle branch block, arrest (rare), hypertension, vascular calcification

Metastatic Calcification

Band keratopathy, red eye syndrome, conjunctival calcification, nephrocalcinosis, vascular calcification, pruritus

Hypercalciuria induced by hypercalcemia causes nephrogenic diabetes insipidus, with polyuria and polydipsia leading to ECF volume depletion and a decreased glomerular filtration rate (GFR) and causing further increases in the serum calcium level. The effect of hypercalcemia on urinary concentration is mediated through CaSR activation, which decreases vasopressin-dependent aquaporin-2 (AQP2) water channels trafficking in the inner medullary collecting duct.¹⁷ Nephrolithiasis and nephrocalcinosis are common complications of hypercalcemia, seen in 15% to 20% of cases of primary HPT, respectively.

LABORATORY FINDINGS

Laboratory findings in patients with hypercalcemia include abnormalities related to the underlying disease causing the hypercalcemia, which are beyond the scope of this chapter. Alterations in the electrocardiogram (ECG) and electroencephalogram (EEG) occur in hypercalcemic patients, independently of the cause of the hypercalcemia. The ECG shows a shortened ST segment and therefore a reduced QT interval as a result of an increased rate of cardiac repolarization. In patients with severe hypercalcemia (>16 mg/dL), there is a widening of the T waves, resulting in an increase in the QT interval. Bradycardia and first-degree heart block may be present in the ECGs of patients with acute and severe

hypercalcemia. The EEG displays slowing and other nonspecific changes.

DIAGNOSIS

A careful history, physical examination, and routine laboratory tests will, in most patients, lead to the correct diagnosis in hypercalcemia. A flow diagram for the evaluation of hypercalcemia is shown in Fig. 18.1. Primary HPT (PHPT) and malignancy-associated hypercalcemia together are responsible for 90% of cases of hypercalcemia, with malignancy being the most common cause in hospitalized patients and PHPT being the most common cause in the outpatient clinic.^{15,18–20}

It is generally easy to differentiate these two entities. Hypercalcemia is only rarely an early finding in occult malignancy. PTH levels are essential in the diagnosis of hypercalcemia. There are two types of assay for PTH, depending on which epitopes of 1-84 PTH are recognized by the antibodies in the assay. Second-generation assays, the immunoradiometric assay (IRMA) and immunochemiluminometric assay (IMCA), use antibodies against amino acid 7-34 and 39-84 epitopes of 1-84 PTH. Despite their being called “intact PTH assays,” implying that they measure only the biologically active PTH, they also detect large carboxy-terminal fragments of PTH, such as PTH (7-84). Thus, they may overestimate the amount of bioactive hormone in serum, especially in CKD patients. The third-generation, whole or bioactive PTH assays, use antibodies against amino acid 1-5 and C-terminal epitopes and detect biologically active intact PTH. The normal levels of PTH measured by various assays range from 8 to 80 Pg/mL (1–9 pmol/L).^{2,21} Intraoperative PTH measurements are frequently used to assess the adequacy of parathyroidectomy. There is no cross-reactivity between PTH and PTHrP assays. In PHPT, levels of PTH can be frankly elevated but can also be in the middle or upper range of normal, particularly in younger individuals (Fig. 18.2).

The differential diagnosis of patients with hypercalcemia and elevated PTH level includes HPT due to thiazide diuretics or lithium, familial hypocalciuric hypercalcemia (FHH), and the tertiary HPT associated with chronic dialysis and kidney transplantation. Patients with FHH have a positive family history, onset of hypercalcemia at a young age, very low urinary calcium excretion, and specific gene abnormalities. In malignancy-associated hypercalcemia and in hypercalcemia of most other causes, PTH levels are low. The diagnosis of humoral hypercalcemia of malignancy (HHM) frequently can be made on clinical grounds. In addition, PTHrP can now be assayed by commercial clinical laboratories to support HHM or when the cause of hypercalcemia is obscure.

Approximately 10% of cases of hypercalcemia are due to other causes. Of particular importance in the evaluation of a hypercalcemic patient are the family history (because of familial syndromes, including multiple endocrine neoplasia type 1 [MEN1], MEN2, and familial hypocalciuric hypercalcemia), medication history (because of the several medication-induced forms of hypercalcemia), and presence of other disease (e.g., granulomatous or malignant disease). Plasma 1,25(OH)₂D levels should be measured when granulomatous disorders or 1,25(OH)₂D lymphoma syndrome is considered. Very high 1,25(OH)₂D levels may suggest vitamin D intoxication as a cause of hypercalcemia. However, this condition is rare.

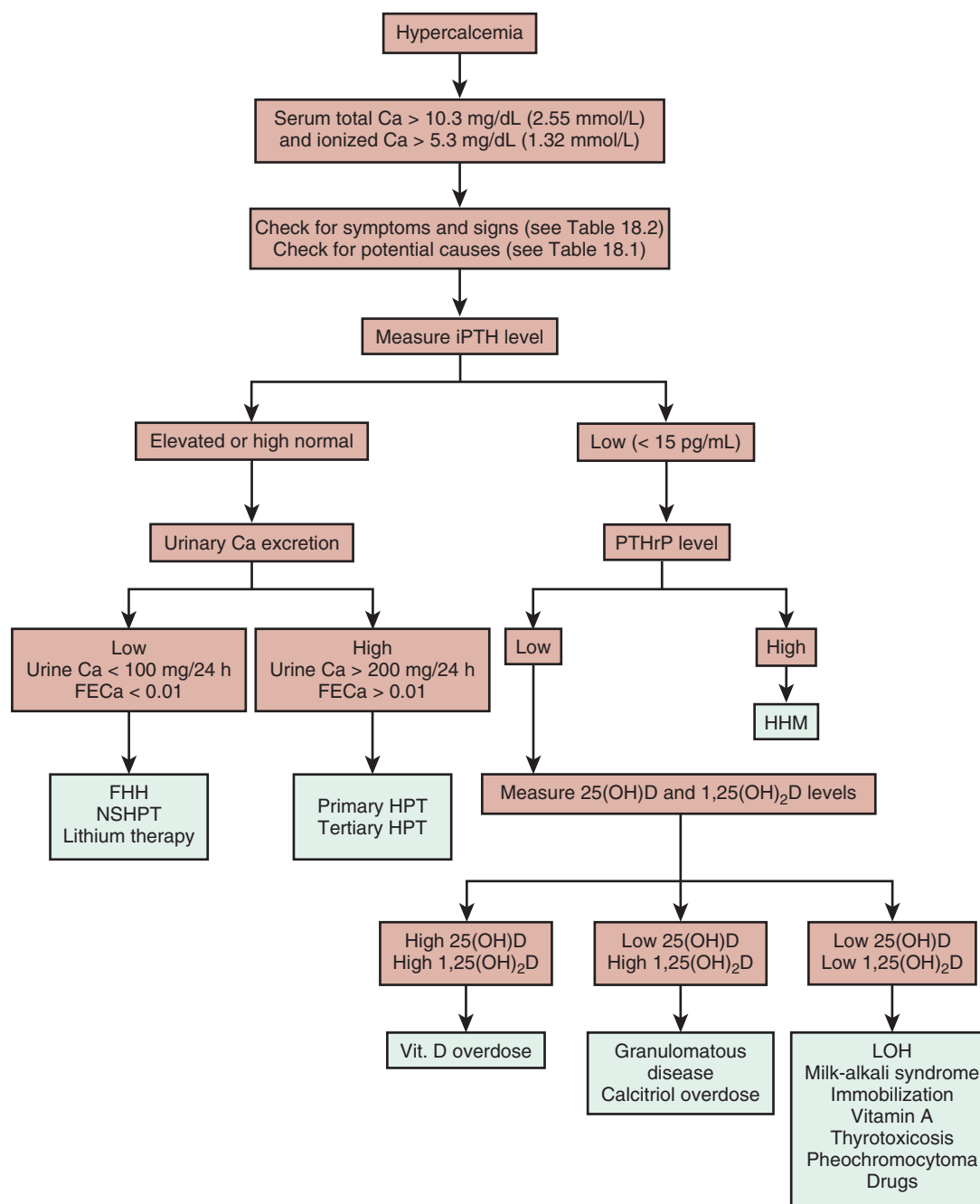


Fig. 18.1 Algorithm for the evaluation of hypercalcemia. *FECa*, Fractional excretion of calcium; *FHH*, familial hypocalciuric hypercalcemia; *HHM*, humoral hypercalcemia of malignancy; *HPT*, hyperparathyroidism; *iPTH*, intact parathyroid hormone; *LOH*, localized osteolytic hypercalcemia; *NSHPT*, neonatal severe hyperparathyroidism; *PTHrP*, parathyroid hormone-related peptide.

CAUSES

Primary Hyperparathyroidism

PHPT is caused by excessive and incompletely regulated secretion of PTH, with consequent hypercalcemia and hypophosphatemia (see Box 18.1). It is the underlying cause of approximately 50% of hypercalcemic cases in the general population. The estimated prevalence of PHPT is about 1%, but may be as high as 2% in postmenopausal women.^{22,23} The annual incidence is approximately 0.03% to 0.04%.^{22–24}

A single enlarged parathyroid gland (adenoma) is the cause of PHPT in 80% to 85% of cases. These adenomas are benign clonal neoplasms of parathyroid chief cells, which lose their normal sensitivity to calcium. In about 15% to 20% of patients with PHPT, all four parathyroid glands are hyperplastic. This occurs in sporadic PHPT or in conjunction with MEN1 or MEN2.¹⁹ In diffuse hyperplasia, the set point for calcium is not changed in any given parathyroid cell, but the increased number of cells causes excess PTH production and hypercalcemia. Parathyroid carcinoma is seen in no more than 0.5% to 1% of patients with PHPT.²⁵

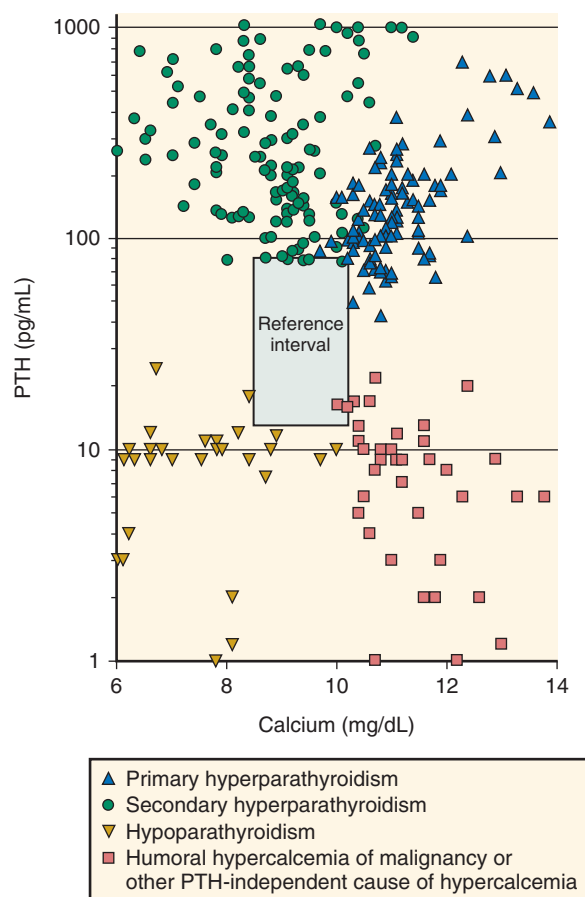


Fig. 18.2 Relationship between total serum calcium and intact parathyroid hormone (PTH) concentrations. Values for patients with known primary hyperparathyroidism, secondary hyperparathyroidism, humoral hypercalcemia of malignancy, or other PTH-independent cause of hypercalcemia, and hypoparathyroidism are plotted. The rectangle represents the normal reference range for the assays. (From O'Neill S, Gordon C, Guo R, et al. Multivariate analysis of clinical, demographic, and laboratory data for classification of patients with disorders of calcium homeostasis. *Am J Clin Pathol.* 2011;135:100–107.)

PHPT occurs at all ages but is most common in older individuals; peak incidence is in the sixth decade of life. After age 50 years, women are about three times more frequently affected than men. Sporadic PHPT is the most common. External neck irradiation during childhood is recognized as a risk factor for PHPT. The genetic alterations underlying parathyroid adenomas have been partially elucidated.²⁶ Rearrangements and overexpression of the PRAD-1–cyclin D1 oncogene have been observed in about 20% of parathyroid adenomas.^{27,28} The MEN1 tumor suppressor gene is inactivated in about 15% of adenomas.^{29,30} Other chromosomal regions may also harbor parathyroid tumor suppressor genes.

PHPT typically presents in one of three ways. In 60% to 80% of cases, there are minimal or no symptoms, and mild hypercalcemia is usually discovered during routine laboratory examination. Another 20% to 25% of patients have a chronic course manifested by mild or intermittent hypercalcemia, recurrent kidney stones, and complications of nephrolithiasis; in these patients, the parathyroid tumor is small (<1.0 g) and slow-growing. In 5% to 10% of patients, there is severe

and symptomatic hypercalcemia and overt osteitis fibrosa cystica; in these patients, the parathyroid tumor is usually large (>5.0 g). Patients with parathyroid carcinoma typically have severe hypercalcemia, with classic kidney and bone involvement.²⁶

The diagnosis of PHPT is now usually suggested by the incidental finding of hypercalcemia rather than by any of the sequelae of PTH excess, such as skeletal and kidney complications or symptomatic hypercalcemia.¹⁸ Hypercalcemia may be mild and intermittent. Hypercalciuria was noted in 40% of PHPT, nephrolithiasis in 19%, and classic bone disease and osteitis fibrosa cystica only in 2% of studies performed between 1984 and 2000 in the United States.¹⁶ However, even in individuals with mild PHPT, there is also progressive bone loss, as measured by bone mineral densitometry over 15 years of observation.³¹

The diagnosis of PHPT is established by laboratory tests showing hypercalcemia, inappropriately normal or elevated blood levels of PTH, hypercalciuria, hypophosphatemia, phosphaturia, and increased urinary excretion of cAMP. Hyperchloremic acidosis may be present, and the ratio of serum chloride to phosphorus is elevated. The serum levels of alkaline phosphatase and uric acid may be also elevated. The serum concentration of magnesium is usually normal but may be low or high.

Some controversy surrounds the potential relationship between PHPT and increased mortality.¹⁵ A number of studies have shown that PHPT may be associated with hypertension, dyslipidemia, diabetes, increased thickness of the carotid artery,³ and increased mortality, primarily from cardiovascular disease.^{20,32} The morbidity from PHPT can also be substantial, especially in symptomatic patients with severe hypercalcemia and a late diagnosis.

The classic bone lesion in PHPT, osteitis fibrosa cystica, is now rarely seen. Diffuse osteopenia is more common.^{20,32} Even in asymptomatic patients, increased rates of bone turnover are always present.^{20,32}

Surgery is still standard therapy for PHPT.^{18,32,33} It is generally agreed that parathyroidectomy is indicated for all patients with biochemically confirmed PHPT who have specific symptoms or signs of disease, such as a history of life-threatening hypercalcemia, CKD, and/or kidney stones. In 2013, a Fourth International Workshop on Hyperparathyroidism updated the guidelines for the management of asymptomatic PHPT.³³ Surgery is advised for asymptomatic disease in patients with serum calcium levels greater than 1 mg/dL above normal, reduced bone mass (T-score < −2.5 at lumbar spine, total hip, femoral neck, or distal third of the radius), creatinine clearance less than 60 mL/min, or age younger than 50 years. Hypercalciuria (>400 mg calcium/24 hours) with increased stone risk by biochemical stone risk analysis and the presence of nephrolithiasis or nephrocalcinosis by imaging techniques are currently considered an indication for parathyroid surgery. Patients older than 50 years with no obvious symptoms should receive close follow-up, including measurements of bone density every 1 to 2 years and serum creatinine and calcium levels annually. All monitored patients should be repleted with vitamin D to achieve a 25-hydroxyvitamin D (25[OH]) D level above 20 ng/dL and should maintain calcium intake the same as individuals without PHPT.

Preoperative localization of the parathyroid glands has generally been considered unnecessary in uncomplicated

patients undergoing surgery for the first time with bilateral neck exploration. However, imaging studies are recommended to be used in complicated cases and if minimally invasive surgery is planned.³⁴ Sestamibi scanning is the most popular and sensitive technique to localize PTH glands, with accuracy rates up to 94%, followed by ultrasound of the neck.^{32,35} If a single adenoma is visualized, minimally invasive parathyroidectomy may be an option, with a cure rate of 95% to 98%: this procedure requires the surgeon to visualize only one gland as long as resection results in a substantial intraoperative decline in PTH level. Otherwise, all four parathyroid glands should be surgically identified. Recurrence of HPT is rare after identification and removal of one enlarged gland.³⁶ If the initial exploration fails, and hypercalcemia persists or recurs, more extensive preoperative parathyroid localization should be performed.^{18,32,35} Complications are greater with reexploration of the neck than after the initial operation.

Although parathyroidectomy remains the definitive treatment of PHPT, patients refusing surgery, those with contraindications for surgery, or those who do not meet current operative guidelines can be treated pharmacologically. There are four classes of medications that can be useful—calcimimetics, bisphosphonates, estrogens, and selective estrogen receptor modulators.^{32,37} There are insufficient long-term data to recommend any of these medications as alternatives to surgery. The CaSR agonist, cinacalcet, has been approved in some European countries for PHPT and in the United States for severe hypercalcemia in adult patients with PHPT who are unable to undergo parathyroidectomy. Cinacalcet therapy in PHPT patients reduced plasma PTH levels, normalized serum calcium levels in the short and long terms, and preserved bone mineral density (BMD).^{37–40} Bisphosphonates and hormone replacement therapy decreased bone turnover and increased BMD in PHPT patients without change in serum calcium levels.³⁷

Parathyroid carcinoma probably accounts for less than 1% of PHPT cases.²⁵ The diagnosis of parathyroid carcinoma may be difficult to make in the absence of metastases because the histologic appearance may be similar to that of atypical adenomas.⁴¹ In general, parathyroid carcinomas are typically large (3 cm), irregular, hard tumors with a low degree of aggressive growth, and survival is common if the entire gland can be removed.^{25,42} Cinacalcet is approved for patients with inoperable parathyroid cancer to control hypercalcemia.

Malignancy

Hypercalcemia occurs in approximately 10% to 25% of patients with some cancer, especially during the last 4 to 6 weeks of their life. It can be classified into four categories—HHM, local osteolytic hypercalcemia (LOH), 1,25 (OH)₂ vitamin D–induced hypercalcemia, and ectopic secretion of authentic PTH.^{43,44}

HHM from secretion of PTHrP by a malignant tumor accounts for approximately 80% of cases. Numerous types of malignancies are associated with HHM, including squamous cell cancer (e.g., head and neck, esophagus, cervix, lung) and renal cell carcinoma, breast cancer, and ovarian carcinoma. Lymphomas associated with human T-lymphotropic virus type 1 (HTLV-1) infection may cause PTHrP-mediated HHM, and other non-Hodgkin lymphomas may also be associated with PTHrP-mediated hypercalcemia.^{44,45} PTHrP

is a large protein encoded by a gene on chromosome 12; it is similar to PTH only at the NH₂ terminus, where the initial eight amino acids are identical.⁴⁵ PTHrP is widely expressed in a variety of tissues, including keratinocytes, mammary gland, placenta, cartilage, nervous system, vascular smooth muscle, and various endocrine sites.⁴⁶ Injection of PTHrP produces hypercalcemia in rats⁴⁷ and essentially reproduces the entire clinical syndrome of HHM, but other circulating factors such as cytokines may also be important. Normal circulating levels of PTHrP are negligible; these are probably unimportant in normal calcium homeostasis. However, mice with a targeted disruption in the PTHrP gene have shown a lethal defect in bone development,^{48–50} thus demonstrating its importance in normal development.

Circulating PTHrP interacts with the PTH/PTHrP receptor in bone and the kidney tubule. It activates bone resorption and suppresses osteoblastic bone formation, thus causing flux of calcium (up to 700–1000 mg/day) from bone into ECF. The reason for this uncoupling of bone formation from bone resorption remains unclear.⁴⁵ One possible explanation is a difference in the affinity of PTH versus that of PTHrP for the PTH receptor on osteoblasts.⁴⁹ PTHrP mimics the anticalciuric effect of PTH on the kidney, which exacerbates hypercalcemia. Other effects of PTHrP include phosphaturia, hypophosphatemia, and increased cAMP excretion by the kidney.

HHM is associated with a reduction in 1,25(OH)₂D levels (in contrast to PHPT), which may limit intestinal calcium absorption. Patients are hypercalcemic and hypophosphatemic and demonstrate increased osteoclastic bone resorption, increased urinary cAMP levels, and hypercalciuria.

Local osteolytic hypercalcemia (LOH) accounts for 20% of patients with malignancy-associated hypercalcemia. LOH-producing tumors include breast and prostate cancers and hematologic neoplasms (e.g., multiple myeloma, lymphoma, leukemia). LOH is caused by locally produced osteoclast-activating cytokines, which include PTHrP, interleukin-1 (IL-1), IL-6, and IL-8. PTHrP increases RANKL expression in osteoblasts and RANK-mediated osteoclast bone resorption. The resorbing bone releases transforming growth factor-β (TGF-β), which in turn stimulates PTHrP expression on tumor cells.^{50,51} The bone metastases can be classified as osteolytic, osteoblastic, or mixed. Osteolytic lesions are caused by osteoclast activation by malignant cells and appear as areas of increased radiolucency on radiographs. LOH leads to predictable pathophysiologic events, which include hypercalcemia, suppression of circulating PTH and 1,25(OH)₂D, hyperphosphatemia, and hypercalciuria. Bone metastases may produce severe pain and pathologic fractures.

Hypercalcemia in breast cancer is associated with the presence of extensive osteolytic metastases and HHM.^{50,51} Extensive osteolytic bone destruction is seen in multiple myeloma.⁵² Although bone lesions develop in all patients with myeloma, hypercalcemia occurs only in 15% to 20% of patients in later stages of disease and with impaired kidney function. The degree of hypercalcemia and bone destruction has not been not well correlated.⁵³ Treatment with bisphosphonates appears to protect against the development of skeletal complications (including hypercalcemia) in patients with myeloma and lytic bone lesions.⁵⁴

Hypercalcemia caused by 1,25(OH)₂ vitamin D production by malignant lymphomas has been reported.^{55,56} All types of

lymphoma can cause this syndrome. The malignant cells or adjacent cells overexpress the enzyme 1 α -hydroxylase, which converts 25(OH)D to 1,25(OH) $_2$ D. Hypercalcemia is mainly secondary to increased intestinal calcium, although decreased kidney clearance and bone resorption may also develop. In addition, increased osteoclastic activity mediated through activation of the RANKL pathway by 1,25(OH) $_2$ D– augmented hypercalcemia. Ectopic production of authentic PTH by a nonparathyroid tumor may occur, but is very rare.⁵⁷

Familial Primary Hyperparathyroidism Syndromes

Familial primary HPT syndromes are defined by a combination of hypercalcemia and elevated or nonsuppressed serum PTH levels.

Familial Hypocalciuric Hypercalcemia and Neonatal Severe Hyperparathyroidism. FHH (benign) is a rare disease (estimated prevalence, 1/78,000), with autosomal dominant inheritance, high penetrance for hypercalcemia, and relative hypocalciuria.^{58–60} FHH was first described in 1966 by Jackson and Boonstra and in 1972 by Foley and colleagues.^{61,62} The hypercalcemia is typically mild to moderate (10.5–12 mg/dL), and affected patients do not exhibit the typical complications associated with elevated serum calcium concentrations. Both total and ionized calcium concentrations are elevated, but the PTH level is generally inappropriately normal, although mild elevations in approximately 15% to 20% of cases have been reported. Urinary calcium excretion is not elevated, as would be expected in hypercalcemia of other causes. The fractional excretion of calcium is usually less than 1%.⁵⁹ The serum magnesium level is commonly mildly elevated, and the serum phosphate level is decreased. Bone mineral density is normal, as are vitamin D levels. The finding of a right-shifted set point for Ca $^{2+}$ -regulated PTH release in FHH has indicated the role of CaSR in FHH.⁶³

Most families have FHH type 1, which is caused by autosomal dominant loss-of-function mutations in the CaSR gene located on chromosome 3q, which encodes for the CaSR. FHH types 2 and 3 are much rarer and are caused by heterozygous mutations in the GNA11 (guanine nucleotide-binding protein alpha-11) gene or the AP2S1 (adaptor-related protein complex 2, sigma 1 subunit) gene, respectively.^{64,65} Both genes localize on chromosome 19q13, and their mutations render the CaSR less sensitive to extracellular calcium.

The fact that relative hypocalciuria persists, even after parathyroidectomy in FHH patients, confirms the role of CaSR in regulating kidney calcium handling.⁶⁶ More than 257 mutations have been described for the CaSR, most of which are inactivating and missense mutations, found throughout the large predicted structure of the CaSR protein.^{63,67,68} Expression studies of mutant CaSRs have shown great variability in their effect on calcium responsiveness. In some cases, CaSR mutations only slightly shift the set point for calcium; other mutations appear to render the receptor largely inactive.^{69–71} CaSR mutation analysis has an occasional role in the diagnosis of FHH in cases in which biochemical test results remain inconclusive, and the distinction of FHH from mild primary HPT is unclear. It is critically important to make an accurate diagnosis because the hypercalcemia in FHH is benign and does not respond to subtotal parathyroidectomy.

Hypercalcemia in FHH has a generally benign course and is resistant to medications, but it has been successfully treated with cinacalcet in some patients.⁷² A potential benefit of calcimimetic agents in FHH has been supported by in vitro studies with human CaSR mutants, which have shown that the calcimimetic agent R-568 enhances the potency of extracellular calcium toward the mutants.⁷³

Patients who inherit two copies of CaSR alleles bearing inactivating mutations develop neonatal severe HPT (NSHPT). NSHPT is an extremely rare disorder that is often reported in the offspring of consanguineous FHH parents; it is characterized by severe hyperparathyroid hyperplasia, PTH elevation, severe hyperparathyroid bone disease, and elevated extracellular calcium levels.^{58,63,74,75} In a few affected infants, only one defective allele has been found, but it is unclear whether this finding is due to the presence of an undetected defect in the other CaSR allele. Treatment is total parathyroidectomy, followed by vitamin D and calcium supplementation. This disease is usually lethal without surgical intervention.

Multiple Endocrine Neoplasia. MEN1 is a rare, autosomal dominant disorder with an estimated prevalence of two to three cases/100,000, characterized by endocrine tumors in at least two of three main tissues—parathyroid gland, pituitary gland, and enteropancreatic tissue. It is the most common form of familial PHPT. PHPT is present in 87% to 97% of patients, whereas pancreatic and pituitary tumors are more likely to be absent.^{18,76} The responsible gene, MEN1,⁷⁷ encodes a nuclear protein, menin, of 610 amino acids, which functions in cell division, genome stability, and transcription regulation.⁷⁸

MEN2A is a syndrome of heritable predisposition to medullary thyroid carcinoma, pheochromocytoma, and PHPT. Mutations in the RET proto-oncogene, which encodes a tyrosine kinase receptor, are responsible for MEN2A.⁷⁹ The biochemical diagnoses and indications for surgery for patients with PHPT in association with MEN1 or MEN2A are similar to those for sporadic PHPT.⁸⁰

Hyperparathyroidism–Jaw Tumor Syndrome.

Hyperparathyroidism–jaw tumor (HPT-JT) syndrome is a rare autosomal dominant disorder characterized by severe hypercalcemia, parathyroid adenoma, and fibro-osseous tumors of mandible or maxilla.⁸¹ Kidney manifestations include cysts, hamartomas, and Wilms' tumor. Mutations in the *HRPT2* (hyperparathyroidism 2) gene, and elimination of its product, parafibromin, which has tumor suppressor activity, are responsible for HPT-JT.⁸² If biochemical changes consistent with PHPT are present, parathyroidectomy is indicated.

Nonparathyroid Endocrinopathies

Hypercalcemia may occur in patients with other endocrine diseases. Mild hypercalcemia is present in up to 20% of patients with hyperthyroidism, but severe hypercalcemia is uncommon.^{83,84} Thyroid hormones (e.g., thyroxine, triiodothyronine) increase bone resorption and lead to hypercalcemia and/or hypercalciuria when bone resorption exceeds bone formation significantly.⁸⁵ Because of a possible increased association between hyperthyroidism and parathyroid adenoma, the latter must be ruled out in patients with thyrotoxicosis and hypercalcemia. Furthermore, the hypercalcemia in a patient

with thyrotoxicosis should be attributed to this disease only if it resolves after achieving an euthyroid state.

Pheochromocytoma may be associated with hypercalcemia⁸⁶; it is usually caused by coincident PHPT and MEN2A. In some patients, hypercalcemia disappears after removal of the adrenal tumor, and some of these tumors produce PTHrP.⁸⁷ Acute adrenal insufficiency is a rare cause of hypercalcemia.⁸⁸ Because these patients may be dehydrated and have hemoconcentration, a rise in the serum albumin concentration and increased binding of calcium to serum albumin secondary to hyponatremia may contribute to the increase in serum calcium levels. In addition, isolated adrenocorticotrophic hormone (ACTH) deficiency can result in hypercalcemia.⁸⁹

Growth hormone administration⁹⁰ and acromegaly⁹¹ have both been associated with hypercalcemia. Acromegaly is often (15%–20% of cases) accompanied by mild hypercalcemia, which results from enhanced intestinal calcium absorption and augmented bone resorption.^{92,93} In acromegalic patients with hypercalcemia, the serum levels of PTH are normal but may be inappropriately high for the levels of serum calcium.

Vitamin D-Mediated Hypercalcemia

Vitamin D is naturally generated in skin under exposure to ultraviolet B (UVB) light or is acquired from the diet and medical supplements. Excess of vitamin D or its metabolites can cause hypercalcemia and hypercalciuria. The mechanism of hypercalcemia is a combination of increased intestinal calcium absorption and bone resorption induced by vitamin D and decreased kidney calcium clearance resulting from dehydration. The effect of toxic amounts of vitamin D is due to an increase in total plasma 25(OH)D, well in excess of 100 ng/mL, which exceeds the binding capacity of vitamin D-binding protein (DBP) for 25(OH)D. The resulting increase in free circulating 25(OH)D may activate the vitamin D nuclear receptor (VDR). Vitamin D metabolites may also displace 1 α ,25(OH)₂D from DBP, increasing free 1 α ,25(OH)₂D levels and thus increasing signal transduction.⁹⁴

Hypercalcemia has been reported in accidental overdoses of vitamin D from fortified cow's milk,^{95,96} consumption by children of a fish oil with manufacturing error that caused an excess of vitamin D, and over-the-counter supplements.⁹⁷ Serum 25(OH)D levels were elevated, 1,25(OH)₂D levels were normal, and PTH levels were depressed or normal in these settings. However, vitamin D well in excess of the tolerable upper intake of 2000 IU/day is required for this form of hypercalcemia to develop.⁹⁸ Polymorphism in genes that regulate vitamin D metabolism may predispose certain individuals to develop toxicity, even with exposure to small vitamin D doses.⁹⁶ The diagnosis is made by the history and detection of elevated 25(OH)D levels. In the syndrome of idiopathic infantile hypercalcemia, a defect in degradation of 1,25(OH)₂D to 24,25(OH)₂D caused by a CYP24A1 loss-of-function mutation is responsible for extremely high levels of 1,25(OH)₂D and hypercalcemia. Vitamin D analogues, including 1,25(OH)₂D used in the treatment of HPT and metabolic bone disease in CKD patients, can also cause hypercalcemia.⁹⁹

Medications

Hypercalcemia and HPT is a long-recognized, well-described consequence of lithium therapy.^{100–103} The prevalence of

lithium-associated hypercalcemia has been estimated to be 4% to 6%.^{101,104} Lithium probably interferes with signal transduction elicited by the CaSR, which increases the set point for extracellular calcium to inhibit PTH secretion.^{103–105} This leads to parathyroid hyperplasia or adenoma.¹⁰⁶ The spectrum of lithium-induced calcium disorders is wide and includes patients with overt HPT and mild or severe hypercalcemia, with or without elevated PTH levels. Hypocalciuria is common, although hypercalciuria was reported in a few case series. Hypercalcemia can be reversible after a few weeks of discontinuing lithium in most patients with short lithium treatment (<5 years). The CaSR agonist cinacalcet was also used with good results in those patients when cessation of lithium therapy was not an option.¹⁰⁷ Symptomatic patients with HPT should be treated with parathyroidectomy.^{102,103}

Vitamin A intake in doses exceeding the recommended daily allowance over prolonged periods, especially in older adults and patients with impaired kidney function, may cause hypercalcemia, with increased alkaline phosphatase levels, presumably from increased osteoclast-mediated bone resorption.^{108–110} The hypercalcemia is accompanied by high retinol plasma levels, and discontinuation of vitamin A caused normalization of plasma calcium levels. Vitamin A analogues, used in the management of dermatologic and hematologic malignant diseases, have also been reported to cause hypercalcemia.^{111,112}

Estrogens and selective estrogen receptor modifiers (e.g., tamoxifen) used in the management of breast cancer may cause hypercalcemia early during treatment, even in the presence of bone metastasis.¹¹³

The incidence of thiazide-associated hypercalcemia increased after 1997 and peaked in 2006, with an annual incidence of 20 cases/100,000, compared with an overall rate of 12 cases/100,000 in 1992–2010 in the population of Olmsted County, Minnesota.^{114,115} A reduction in urinary calcium excretion, volume contraction, and metabolic alkalosis are the major reasons for thiazide-induced hypercalcemia. Also, thiazides may increase intestinal calcium absorption and reveal PHPT.^{115–117} Hypercalcemia is usually mild and asymptomatic. Subjects with unsuppressed PTH levels, despite hypercalcemia or with severe and continued hypercalcemia, may have PHPT.

Many other medications occasionally cause hypercalcemia, including theophylline, foscarnet, growth hormone, parenteral nutrition, and manganese in toxic doses.

Milk-Alkali Syndrome

Milk-alkali syndrome was originally described in patients with duodenal ulcers receiving therapy with sodium bicarbonate and large amounts of milk. These patients have hypercalcemia, hyperphosphatemia, hypocalciuria, and CKD, together with kidney and other soft tissue calcifications.¹¹⁸ During the last 20 years, calcium supplements in the form of calcium carbonate for the prevention and treatment of osteoporosis have become the main cause of this syndrome.^{119,120} In the most recent literature, the name “calcium-alkali syndrome” was proposed.¹²¹ In some studies, milk-alkali syndrome was the third most common cause of hypercalcemia in non-end-stage kidney disease (ESKD) hospitalized patients.¹²² The pathogenesis of milk-alkali syndrome can be divided into two phases, the generation of hypercalcemia by the intake of calcium and the maintenance phase. Usually, oral intake of more

than 4g of elemental calcium/day has been reported, but even 2 g of calcium/day, especially if taken together with vitamin D, may induce this syndrome. Hypercalcemia activates the kidney CaSR, causing natriuresis and water diuresis, with volume depletion and a decrease in the GFR. Increased tubular reabsorption of calcium as a result of metabolic alkalosis and volume depletion contribute to the maintenance of hypercalcemia.¹²⁰ The diagnosis is made largely by the history and may not be obvious because of atypical dietary sources of calcium and alkali. Hypercalcemia can be corrected, but kidney damage may be permanent.

Immobilization

Immobilization, especially in high bone turnover states (e.g., in young people and in those with hyperparathyroidism, breast cancer with bone involvement, Paget disease), suppresses osteoblastic bone formation and increases osteoclastic bone resorption, leading to uncoupling of these two processes, with subsequent release of calcium from the bone and hypercalcemia.^{123,124} Typically, it takes from 10 days to a few weeks for the development of immobilization hypercalcemia. Increased sclerostin production by osteocytes during mechanical unloading and disuse of the bone is implicated in the pathogenesis of hypercalcemia.¹²⁵ Sclerostin is a glycoprotein that inhibits Wnt/ β catenin signaling in the osteoblast and decreases bone formation.¹²⁶ It is of interest that antisclerostin antibodies are being studied for the treatment of osteoporosis. Bisphosphonates may help decrease hypercalcemia and osteopenia in the setting of immobilization-induced hypercalcemia.¹²⁷ There are case reports showing that denosumab also can correct hypercalcemia of immobilization. Denosumab is a fully humanized monoclonal antibody that prevents the binding of RANK to RANKL, thereby reducing the formation, function, and survival of osteoclasts. Mobilization remains the ultimate cure for immobilization-associated hypercalcemia.

Granulomatous Disease

A variety of granulomatous diseases are associated with hypercalcemia. The most common is sarcoidosis—prevalence of hypercalcemia and hypercalciuria of 10% and 20%, respectively—but tuberculosis, berylliosis, histoplasmosis, coccidioidomycosis, pneumocystosis, leprosy, histiocytosis X, eosinophilic granulomatosis, and inflammatory bowel disease may present with hypercalcemia.^{89,128–130} Hypercalcemia is more common in chronic and disseminated granulomatous diseases. Sun exposure, or even small doses of vitamin D supplementation, may precipitate or worsen this syndrome. The hypercalcemia, which has been best studied in sarcoidosis, is caused by inappropriate extrarenal production of 1,25(OH)₂D by activated macrophages with increased 1 α -hydroxylase activity.^{131,132} Elevated circulating 1,25(OH)₂D levels have been described in most granulomatous diseases during hypercalcemia, except in coccidioidomycosis. The 1,25(OH)₂D in turn leads to intestinal hyperabsorption of calcium, hypercalciuria, and hypercalcemia. Osteopontin, highly expressed by histiocytes in granulomas, may contribute to hypercalcemia via osteoclast activation and bone resorption. Bone mineral content tends to be reduced in these patients. Hypercalciuria may precede hypercalcemia and may be an early indicator of this complication.

Standard treatment consists of administration of glucocorticoids, which decreases the abnormal 1,25(OH)₂D produc-

tion.¹³³ Chloroquine and ketoconazole, which also decrease 1,25(OH)₂D production by competitive inhibition of CYP450-dependent 1 α -hydroxylase, have also been shown to be efficacious.^{134,135}

Liver Disease

Hypercalcemia has been reported in patients with end-stage liver disease with hyperbilirubinemia who are awaiting liver transplantation in the absence of HPT or hypervitaminosis D.¹³⁶

Acute and Chronic Kidney Disease

Hypercalcemia may be observed in patients with certain forms of acute and chronic kidney disease and in kidney transplant recipients. These conditions are discussed in detail in Chapter 55.

MANAGEMENT OF HYPERCALCEMIA

The optimal therapy for hypercalcemia must be tailored to the severity of hypercalcemia, clinical condition, and underlying cause (Table 18.1).¹⁵ Theoretically, a decrease in serum calcium levels can be achieved by enhancing its urinary excretion, augmenting net movement of calcium into bone, inhibiting bone resorption, reducing intestinal absorption of calcium, and/or removing calcium from the ECF by other means. Patients with mild hypercalcemia (<12 mg/dL) do not require immediate treatment. They should discontinue any medications implicated in causing hypercalcemia, avoid volume depletion and physical inactivity, and maintain adequate hydration. Moderate hypercalcemia (12–14 mg/dL), especially if acute and symptomatic, requires more aggressive therapy. Patients with severe hypercalcemia (>14 mg/dL), even without symptoms, should be treated intensively.

Volume Repletion and Loop Diuretics

Correction of the ECF volume is the first and most important step in the treatment of severe hypercalcemia from any causes. It can be achieved with a normal isotonic saline infusion at 200 to 500 mL/hour, adjusted to obtain a urine output of 150 to 200 mL/hour and with appropriate hemodynamic monitoring.^{15,137,138} Volume repletion can lower the calcium concentration by approximately 1 to 3 mg/dL by increasing GFR and decreasing sodium and calcium reabsorption in the proximal and distal tubules.

Once volume expansion is achieved, loop diuretics can be given concurrently with saline to increase the calciuresis by blocking the Na⁺-K⁺-2Cl⁻ cotransporter in the TAL.¹³⁷ Usually, furosemide is given at a dose of 40 to 80 mg every 6 hours and this, together with saline therapy, may decrease the serum calcium concentration by 2 to 4 mg/dL. Urinary losses of fluid, potassium, and magnesium should be evaluated at intervals of 2 to 4 hours and quantitatively replaced to prevent dehydration, hypokalemia, and hypomagnesemia. Usually, 20 to 40 mEq of KCl and 15 to 30 mg of magnesium ion/L of saline infusion are adequate to replenish the urinary losses of these electrolytes. Care must be taken to monitor the patient's volume status closely during the administration of large amounts of saline and diuretic, particularly in hospitalized patients with cardiac or pulmonary disease. It must be noted that the use of loop diuretics for hypercalcemia is not supported by any randomized controlled studies and has

Table 18.1 Pharmacologic Therapy for Hypercalcemia^a

Intervention	Dose	Adverse Effects
Hydration or Calciuresis		
Intravenous saline	200–500 mL/h, depending on patient's cardiovascular and kidney status	Congestive heart failure
Furosemide	20–40 mg IV (after rehydration has been achieved)	Dehydration, hypokalemia, hypomagnesemia
First-Line Medications		
IV bisphosphonates ^b		
Pamidronate	60–90 mg IV over 2 hours in 50–200 mL saline solution or 5% dextrose in water ^d	Acute kidney injury, transient flulike syndrome with aches, chills, and fever
Zoledronate	4 mg IV over 15 minutes in 50 mL of saline solution or 5% dextrose in water	Acute kidney injury, transient flulike syndrome with aches, chills, and fever
Second-Line Medications		
Glucocorticoids ^c	Example: prednisone, 60 mg orally daily, for 10 days	Potential interference with chemotherapy; hypokalemia, hyperglycemia, hypertension, Cushing syndrome, immunosuppression
Mithramycin	Single dose of 25 µg/kg of body weight over 4–6 hours in saline	Thrombocytopenia, platelet aggregation defect, anemia, leukopenia, hepatitis, kidney failure ^e
Calcitonin	4–8 U/kg subcutaneously or intramuscularly every 12 hours	Flushing, nausea, escape phenomenon
Gallium nitrate	100–200 mg/m ² of body surface area IV given continuously over 24 hours for 5 days	Acute kidney injury
Denosumab ^f	120 mg on days 1, 8, 15, and 29 and every 4 weeks	Hypocalcemia, hypophosphatemia, osteonecrosis of the jaw, atypical femoral fractures

^aMany recommendations in this table are based on historical precedent and common practice rather than on randomized clinical trials. There are data from randomized trials comparing bisphosphonates to the other agents listed and to one another.

^bPamidronate and zoledronate are approved by the US Food and Drug Administration (FDA). Ibandronate and clodronate are available in continental Europe, the United Kingdom, and elsewhere. Bisphosphonates should be used with caution, if at all, when the serum creatinine level exceeds 2.5 to 3.0 mg/dL (221.0–265.2 µmol/L).

^cPamidronate is generally used at a dose of 90 mg, but the 60-mg dose may be used to treat patients of small stature or those with kidney impairment or mild hypercalcemia.

^dThese drugs have a slow onset of action, as compared with bisphosphonates; approximately 4 to 10 days are required for a response.

^eThese effects have been reported in association with higher dose regimens used to treat testicular cancer (50 µg/kg body weight/day over a period of 5 days) and in patients receiving multiple doses of 25 µg/kg; they are not expected to occur with a single dose of 25 µg/kg unless preexisting liver, kidney, or hematologic disease is present.

^fApproved by the FDA for the prevention of skeletal-related events in patients with bone metastasis from solid tumor; used in an open-label fashion as a rescue therapy if bisphosphonates are not effective.

Modified from Stewart AF. Clinical practice. Hypercalcemia associated with cancer. *N Engl J Med*. 2005;352:373–379. With permission.

been criticized for this reason.¹³⁸ However, in our opinion, loop diuretics still remain an important tool in the management of hypercalcemia, especially for patients at risk of volume overload.

Inhibition of Bone Resorption

The increase in bone resorption, as the most common pathology leading to hypercalcemia, must be addressed concurrently with volume expansion and hydration. Bisphosphonates are currently the agents of choice in the treatment of mild to severe hypercalcemia, especially that associated with cancer and vitamin D toxicity.⁹⁶ They are pyrophosphate analogues with a high affinity for hydroxyapatite and inhibit osteoclast function in areas of high bone turnover.¹⁵ The US Food and Drug Administration (FDA) has approved two bisphosphonates for the treatment of hypercalcemia, zoledronate (4 mg intravenous [IV] over 15 minutes or longer) and pamidronate (60–90 mg IV over 2–24 hours). The clinical

response takes 48 to 96 hours and is sustained for up to 3 weeks. Doses can be repeated no sooner than every 7 days. Both agents are effective in lowering calcium levels. Zoledronate was slightly more efficacious than pamidronate in a randomized clinical trial.¹³⁹ In Europe, other bisphosphonates, such as clodronate and ibandronate, have also been approved.

Fever is observed in about 20% of patients taking bisphosphonates; rare side effects include acute kidney injury, collapsing glomerulopathy, and osteonecrosis of the jaw. Ibandronate seems to have minimal to no kidney toxicity. The dose of bisphosphonates should be adjusted in patients with preexisting kidney disease.¹⁴⁰ The kidney component of hypercalcemia, which includes increased distal tubular calcium reabsorption driven by PTH-PTHrP, does not respond to bisphosphonates.

Calcitonin is also an effective inhibitor of osteoclast bone resorption. It has a rapid onset (within 12 hours), its effect

is transient, and it has minimal toxicity.^{141–145} Calcitonin is usually given as 4 to 8 U/kg subcutaneously every 6 to 12 hours.^{141,142} Its role is mainly in the initial treatment of severe hypercalcemia while waiting for the more sustained effect of bisphosphonates.

Gallium nitrate inhibits bone resorption by increasing the solubility of hydroxyapatite crystals. The usual dose is 200 mg/m² IV over 24 hours, with adequate hydration for 5 consecutive days; the hypocalcemic effect is not generally observed until the end of this period. Gallium nitrate is effective, but can be nephrotoxic.^{143,144}

Denosumab, is a fully humanized monoclonal antibody that binds to RANKL and inhibits osteoclasts. Denosumab inhibits the maturation of preosteoclasts to osteoclasts by binding to and inhibiting RANKL. In 2010, denosumab was approved by the FDA for postmenopausal women at risk for osteoporosis.¹⁴⁵ It was also approved for the prevention of skeleton-related events in patients with bone metastasis from a solid tumor. It has also been approved for the treatment of hypercalcemia of malignancy not corrected by bisphosphonates.^{145,146}

Plicamycin (Mithramycin), 25 µg/kg IV, every 5 to 7 days, may be used in patients with advanced kidney disease. Other therapies for hypercalcemia, such as chelation with ethylenediaminetetraacetic acid (EDTA) and IV phosphate, have adverse side effect profiles and are no longer recommended.

Glucocorticoids are useful therapy for hypercalcemia in a specific subset of causes. They are most effective in hematologic malignancies (e.g., multiple myeloma, Hodgkin disease) and disorders of vitamin D metabolism (e.g., granulomatous disease, vitamin D toxicity).^{128,133}

In severely hypercalcemic patients who are comatose with changes in the ECG, in severe acute kidney injury, or who cannot receive aggressive hydration, hemodialysis with a low- or no-calcium dialysate is an effective treatment.¹⁴⁷ Continuous kidney replacement therapy can also be used to treat severe hypercalcemia.¹⁴⁸ The effect of dialysis is transitory and needs to be followed by other measures.

As discussed previously, cinacalcet, an allosteric activator of CaSR, is approved for patients with inoperable parathyroid cancer to control hypercalcemia. The off-label use of cinacalcet has been reported in patients with PHPT who have mild disease, failed parathyroid surgery, or contraindications to surgery.^{32,37,39,40} Other hypercalcemic disorders, such as FHH⁷² and lithium-induced HPT, have also been treated with cinacalcet.^{104,107}

HYPOCALCEMIA

Hypocalcemia is usually defined as a total serum calcium concentration, corrected for protein, of less than 8.4 mg/dL and/or an ionized calcium level less 1.16 mmol/L, although these values may vary slightly, depending on the laboratory. Estimation of the ionized calcium concentration based on the total serum calcium corrected for albumin is encumbered with errors, as discussed in the introduction. Thus, ionized calcium should be directly measured before a major workup for the causes of hypocalcemia is undertaken.

Hypocalcemia is highly prevalent in hospitalized patients (10%–18%) and is particularly common in the intensive care unit (70%–80%).^{149,150}

SIGNS AND SYMPTOMS

Acute hypocalcemia can result in severe clinical symptoms that need rapid correction, whereas chronic hypocalcemia may be an asymptomatic laboratory finding. The clinical features of hypocalcemia are summarized in Box 18.3. Their presentation reflects the absolute calcium concentration and the rapidity of its fall. The threshold for overt symptoms depends also on serum pH and the severity of any concurrent hypomagnesemia, hyponatremia, or hypokalemia. The classic symptoms of hypocalcemia include neuromuscular excitability in the form of numbness, circumoral tingling, feeling of pins and needles in the feet and hands, muscle cramps, carpopedal spasms, laryngeal stridor, and frank tetany. Tapping over the facial nerve anterior to the ear can induce facial muscle spasm (Chvostek sign). However, a Chvostek sign may occur in 10% of normal people, and it was negative in 29% of patients with mild hypocalcemia. A Trousseau sign of *main d'accoucheur*, (French for “hand of the obstetrician”), elicited by inflation of a sphygmomanometer cuff placed on the upper arm to 10 mm Hg above systolic blood pressure for 3 minutes, has greater than 90% sensitivity and specificity.¹⁵¹ Patients with hypocalcemia may experience emotional disturbances, irritability, impairment of memory, confusion, delusion, hallucination, paranoia, and depression. Epileptic seizures, often Jacksonian, may occur but are usually not associated with aura, loss of consciousness, and incontinence. Patients with chronic hypocalcemia, including those with idiopathic and postsurgical hypoparathyroidism and those with pseudohypoparathyroidism, may have papilledema, elevated cerebrospinal fluid pressure, and neurologic signs simulating those of a cerebral tumor.

Box 18.3 Clinical Features of Hypocalcemia

Neuromuscular Irritability

- General fatigability and muscle weakness
- Paresthesia, numbness
- Circumoral and peripheral extremity tingling
- Muscle twitching and cramping
- Tetany, carpopedal spasms
- Chvostek sign, Trousseau sign
- Laryngeal and bronchial spasms

Altered Central Nervous System Function

- Emotional disturbances—irritability, depression
- Altered mental status, coma
- Tonic-clonic seizures
- Papilledema, pseudotumor cerebri
- Cerebral calcifications

Cardiovascular

- Lengthening of the QTc interval
- Dysrhythmias
- Hypotension
- Congestive heart failure

Dermatologic and Ocular

- Dry skin, coarse hair, brittle nails
- Cataracts

Bilateral cataracts affecting the anterior and posterior subcapsular areas of the cortical portions of the lens may develop after 1 year of hypocalcemia. The cataracts do not resolve after correction of the hypocalcemia. In patients with idiopathic hypoparathyroidism, the skin could be dry and scaly, eczema and psoriasis may worsen, and candidiasis can occur. The eyelashes and eyebrows may be scanty, and axillary and pubic hair may be absent. Because some forms of this disease have an autoimmune cause, manifestations of other autoimmune diseases, such as adrenal, thyroid, and gonadal insufficiency, diabetes mellitus, pernicious anemia, vitiligo, and alopecia areata may be present and should be sought.

Long-lasting hypocalcemia in children and adults can result in congestive heart failure caused by cardiomyopathy, which is reversible with correction of the calcium.^{152–154} Prolongation of the QTc interval on the ECG is a well-known effect of hypocalcemia on heart conduction.

Hypoparathyroidism in children often causes teeth abnormalities, such as defective enamel and root formation, dental hypoplasia, or failure of adult teeth to erupt. Severe skeletal mineralization may occur in the fetus of untreated pregnant women with hypoparathyroidism and hypocalcemia.

LABORATORY FINDINGS

It is important to establish the diagnosis of hypocalcemia, based not only on the measurement of total calcium with proper adjustment to albumin and pH levels, but with evidence that ionized calcium is low. The alterations in serum PTH and serum and urinary electrolyte levels in various hypocalcemic states depend on the mechanisms responsible for the hypocalcemia (see Fig. 18.2), and knowledge of these changes aids in the differential diagnosis of these disorders.

An x-ray examination of the skull or computed tomography (CT) scanning of the brain may reveal intracranial calcifications, especially of the basal ganglia.¹⁵⁵ These have been noted in up to 20% of hypocalcemic patients with idiopathic hypoparathyroidism but are less common in postsurgical hypoparathyroidism unless the disease is long-standing. Such calcifications are also encountered in patients with pseudohypoparathyroidism. Bone disease may be observed, but its findings differ in the various causes of hypocalcemia (see later).

DIAGNOSIS

The most common causes of hypocalcemia in the nonacute setting are hypoparathyroidism, hypomagnesemia, CKD, and vitamin D deficiencies (Box 18.4). These entities should be considered early in the diagnosis of hypocalcemic individuals. It is conceptually and clinically useful to subclassify hypocalcemic individuals into those with elevated PTH levels and those with subnormal or inappropriately normal PTH concentrations, as in primary hypoparathyroidism (Fig. 18.3). A thorough medical history and physical examination are diagnostically important because hypocalcemia can be caused by postsurgical, pharmacologic, inherited, developmental, and nutritional problems, in addition to being part of complex syndromes.

CAUSES

The causes of hypocalcemia are summarized in Box 18.4. They can be broadly classified into one of three

Box 18.4 Causes of Hypocalcemia

- Inherited and genetic syndromes with hypoparathyroidism
 - PTH gene mutations, isolated congenital hypoparathyroidism
 - Autosomal dominant hypoparathyroidism with activating mutation of the CaSR (OMIM 146200)
 - DiGeorge syndrome (OMIM 188400)
 - Other forms of familial hypoparathyroidism
- Inherited and genetic syndromes with resistance to PTH action
 - Pseudohypoparathyroidism, types 1a, 1b, and 2
 - Hypomagnesemic syndromes
- Acquired hypoparathyroidism, inadequate PTH production
 - Damage or destruction of the parathyroid glands
 - Postsurgical
 - Autoimmune—isolated or with multiple endocrine dysfunction
 - Acquired antibodies against CaSR
 - Polyglandular failure syndrome type I (OMIM 240300 and 607358)
 - Irradiation
 - Metastatic and infiltrative diseases
 - Deposition of heavy metals—iron overload, copper overload
 - Reversible impairment of PTH secretion
 - Severe hypomagnesemia
 - Hypermagnesemia
- Inadequate vitamin D production
 - Vitamin D deficiency—nutritional, lack of sunlight exposure
 - Malabsorption
 - End-stage liver disease and cirrhosis
 - Chronic kidney disease
- Vitamin D resistance
 - Pseudovitamin D deficiency rickets (vitamin D–dependent rickets type 1)
 - Vitamin D–resistant rickets (vitamin D–dependent rickets type 2)
- Miscellaneous causes
 - Hyperphosphatemia
 - Phosphate retention caused by acute or chronic kidney disease
 - Excess phosphate absorption caused by enemas, oral supplements
 - Massive phosphate release caused by tumor lysis or crush injury
- Drugs
 - Foscarnet
 - Bisphosphonate therapy (especially in patients with vitamin D deficiency)
 - Denosumab
 - Rapid transfusion of large volumes of citrate-containing blood
 - Acute critical illness (multiple contributing causes)
 - Hungry bone syndrome, recalcification tetany
 - Postthyroidectomy for Graves disease
 - Postparathyroidectomy
 - Osteoblastic metastases
 - Acute pancreatitis
 - Rhabdomyolysis
 - Substances interfering with the laboratory assay for total calcium
 - Gadolinium salts in contrast agents given during MRI, MRA

CaSR, Calcium-sensing receptor; OMIM, Online Mendelian Inheritance in Man; MRI, magnetic resonance imaging; MRA, magnetic resonance angiography; PTH, parathyroid hormone.

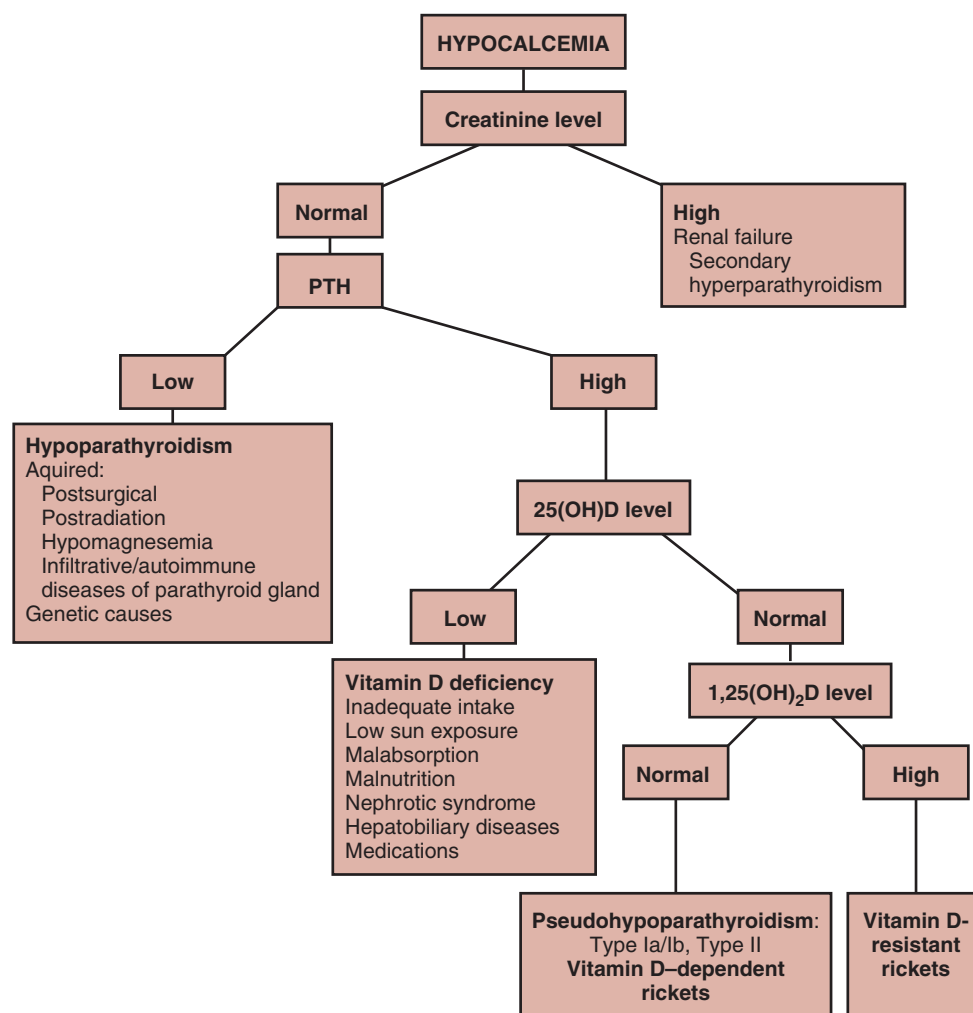


Fig. 18.3 Algorithm for the evaluation of hypocalcemia. PTH, Parathyroid hormone.

categories—PTH-related (hypoparathyroidism and pseudohypoparathyroidism), vitamin D-related (low production, vitamin D resistance), and miscellaneous causes.

Parathyroid Hormone–Related Disorders: Hypoparathyroidism and Pseudohypoparathyroidism

This group of disorders presents with hypocalcemia and hyperphosphatemia caused by failure of the parathyroid gland to secrete adequate amounts of biologically active PTH or resistance to PTH action at the tissue level. Both can be inherited or acquired. The levels of PTH are low or absent in hypoparathyroidism (HP) due to lack of PTH production, but elevated in pseudohypoparathyroidism (PHP) due to a secondary or adaptive increase in PTH secretion. The fractional calcium excretion is elevated in HP and low in PHP. Insufficient $1,25(\text{OH})_2\text{D}$ is generated for efficient intestinal calcium absorption because of decreased activity of 1α -hydroxylase in the proximal tubules. Skeletal response in both categories is appropriate to the levels of circulating PTH, with low bone turnover in HP and excessive bone remodeling in PHP.^{156,157} Hypoparathyroidism is a rare disorder. One study from Japan has found the prevalence to be 7.2 cases/million people.¹⁵⁸

Genetic Causes of Hypoparathyroidism. At least four different mutations affecting the PTH gene(s) involved in parathyroid development have been identified as a cause of familial isolated HP (Table 18.2). All these conditions present during the neonatal period with severe hypocalcemia without any other organ involvement and respond well to therapy with vitamin D analogues.

Heterozygous gain-of-function mutations in the CaSR can activate CaSR or cause CaSR to be hyperresponsive to extracellular calcium.¹⁶³ The phenotype seen is essentially the opposite of FHH and has been termed “autosomal dominant hypoparathyroidism” or “familial hypocalcemia with hypercalciuria.” Patients present with mild hypocalcemia, hypomagnesemia and hypercalciuria, with low or inappropriately normal PTH levels. The set point for PTH secretion is shifted to the left. Treatment with calcium supplements and vitamin D is warranted only for patients with severe symptomatic hypocalcemia. The goal should be to increase the calcium level to render the patient asymptomatic, not necessarily to a normocalcemic level. Kidney calcium excretion requires monitoring because these patients may develop frank hypercalciuria and nephrocalcinosis.^{168,169} Thiazide diuretics or injectable PTH can be used to decrease calciuria at any given level of serum calcium.^{170,171}

Table 18.2 Genetic Syndromes With Hypoparathyroidism

Reference	Syndrome	Other Clinical Features	Process Affected	Inheritance	Gene Mutated	Syndrome OMIM No.
Ding et al ¹⁵⁹	Familial isolated hypoparathyroidism	None	Parathyroid gland development	AR	<i>GCM2</i>	146200
Bowl et al ¹⁶⁰ Parkinson and Thakker ¹⁶¹			PTH gene mutation affecting its synthesis	X-linked AR	<i>SOX3?</i> Prepro-PTH splice site	307700
Arnold et al ¹⁶²				AD	Prepro-PTH signal peptide	
Pollak et al ¹⁶³	Autosomal dominant hypoparathyroidism	Hypocalcemia, hypomagnesemia, hypercalciuria	Calcium sensing	AD	<i>CaSR</i>	146200 601298
Yagi et al ¹⁶⁴	DiGeorge, CATCH22	Cardiac anomalies, abnormal facies, thymic aplasia, cleft palate	Defective third and fourth branchial pouch development	Sporadic or AD	Chromosome 22q11 deletions (including <i>TBX1</i>)	188400
Van Esch et al ¹⁶⁵	HDR	Hypoparathyroidism, deafness, kidney anomalies	Parathyroid development	AD	<i>GATA3</i> transcription factor	146255
Parvari et al ¹⁶⁶	Kenny-Caffey, Sanjad-Sakati	Microcephaly, mental retardation, growth failure ± osteosclerosis		AR	<i>TBCE</i> (chaperone for tubulin folding)	241410 244460
Neufeld et al ¹⁶⁷	Autoimmune polyendocrinopathy—candidiasis—ectodermal dystrophy (APECED)	Chronic mucocutaneous candidiasis, Addison disease	Immune tolerance	AR	<i>AIRE</i> (autoimmune transcriptional regulator)	240300

AD, Autosomal dominant; AR, autosomal recessive; *CaSR*, calcium-sensing receptor; OMIM, Online Mendelian Inheritance in Man; PTH, parathyroid hormone.

A number of rare congenital syndromes with multiple developmental abnormalities can also be associated with familial hypoparathyroidism, including DiGeorge syndrome, hypoparathyroid, deafness, and renal anomalies (HDR), autoimmune polyendocrinopathy—candidiasis—ectodermal dystrophy (APECED), and mitochondrial disorders (see Table 18.2).

Genetic Syndromes With Resistance to Parathyroid Hormone Action. Individuals with PHP are hypocalcemic and hyperphosphatemic but have elevated PTH levels. This condition, reported in 1942 by Albright, was the first described example of a hormone resistance disease.¹⁷² The patients exhibited a pattern of features of Albright hereditary osteodystrophy (AHO) that included short stature, round face, mental retardation, brachydactyly, and the lack of a phosphaturic response to parathyroid extract. PHP is now recognized as a heterogeneous group of related disorders.^{173,174} It may be inherited or sporadic.

PHP is subdivided into two types, depending on the kidney tubular response to infused exogenous PTH. PHP type 1 (PHP-1) refers to complete resistance to the effects of PTH, as demonstrated by the failure of patients to increase serum calcium, urinary cAMP, and phosphate levels in response to PTH infusion.^{175,176} PHP-1 is subdivided into PHP type 1a (PHP-1a), with AHO, and PHP type 1b (PHP-1b), without

AHO. The presence of AHO without hypocalcemia and endocrine dysfunction has been termed “pseudopseudohypoparathyroidism” (PPHP).

PHP-1a and PPHP result from loss-of-function mutations of the *GNAS1* gene, which encodes the stimulatory G protein α -subunit ($G_{\alpha s}$) that couples the type 1 PTH/PTHrP receptor (PTH1R) to the adenylyl cyclase pathway.¹⁷⁷ Patients with the *GNAS1* gene mutation also have resistance to thyroid-stimulating hormone (TSH), gonadotropins, glucagons, calcitonin, and gonadotropin-releasing hormone (GnRH) because the same $G_{\alpha s}$ pathway is used by these hormones. Promoter-specific genomic imprinting of *GNAS1* has been established and provides the probable explanation for the complex phenotypic expression of the dominantly inherited genetic defect. Maternal transmission of the mutation causes PHP-1a; paternal transmission leads to PPHP.¹⁷⁸

PHP-1b appears to be caused by mutations that affect the regulatory elements of *GNAS1*, mainly in the proximal tubules.^{179,180} Patients with PHP-1c exhibit the features of PHP-1, but without defective $G_{\alpha s}$ activity.

PHP-2 is a heterogeneous group of disorders characterized by a reduced phosphaturic response to PTH but a normal increase in urinary cAMP levels.¹⁸¹ The cause is unclear but may be caused by a defect in the intracellular response to cAMP or some other component of the PTH signaling pathway. It does not appear to follow a clear familial pattern.

Hypomagnesemic Syndromes

Impaired PTH secretion and inadequate PTH response to hypocalcemia are typically observed in hypomagnesemic patients. This is corrected by magnesium replacement. Congenital defects leading to hypomagnesemia and hypocalcemia are discussed later (see “Magnesium Disorders”) and in Chapters 44 and 73.

Acquired Hypoparathyroidism and Inadequate Parathyroid Hormone Production

Postsurgical Causes. The most common cause of acquired hypoparathyroidism in adults is surgical removal of or damage to the parathyroid glands. Transient hypocalcemia after thyroid surgery has been observed in 2% to 23% of cases, whereas permanent hypocalcemia occurred in approximately 1% to 2% of cases.^{171,182–184} Hypocalcemia was more likely to occur after total thyroidectomy for cancer, Graves’ disease, radical neck dissection for other cancers, and repeated operations for parathyroid adenoma removal. Hypoparathyroidism may result from inadvertent removal of the parathyroids, damage from bleeding, or devascularization-infarction. Removal of a single hyperfunctioning parathyroid adenoma can result in transient hypocalcemia because of hypercalcemia-induced suppression of PTH secretion from the normal glands. Surgical experience and use of appropriate surgical technique may reduce the frequency of hypothyroidism.¹⁸⁴

The so-called hungry bone or recalcification syndrome represents an important cause of prolonged hypocalcemia after parathyroidectomy or thyroidectomy for any form of HPT or hyperthyroidism, respectively.¹⁸⁵ Postoperative withdrawal of PTH decreases osteoclastic bone resorption, presumably mediated by decreased RANKL release from osteoblasts, without affecting osteoblastic bone formation activity, which leads to increased bone uptake of calcium, phosphate, and magnesium. Risk factors for the development of this hungry bone syndrome include large parathyroid adenomas, age older than 60 years, and high preoperative levels of serum PTH, calcium, and alkaline phosphatase. There have been reports that bisphosphonate therapy for Paget disease and cinacalcet use for secondary HPT can also cause a medical hungry bone syndrome.^{186,187}

Acquired HP from nonsurgical causes is rare, with the exception of autoimmune disorders and magnesium deficiency. Although metal overload diseases (e.g., hemochromatosis, Wilson disease),^{188,189} granulomatous diseases, miliary tuberculosis, amyloidosis, or neoplastic infiltrate are often mentioned as causes of hypoparathyroidism, these entities are rare. Alcohol consumption has been reported to cause transient hypocalcemia.¹⁹⁰

Magnesium Disorders. Both magnesium excess and deficiency can produce generally mild hypocalcemia and reversible HP. Acute infusion of magnesium or hypermagnesemia inhibits PTH secretion.¹⁹¹ Magnesium is an extracellular CaSR agonist, although less potent than calcium. Hypermagnesemia, when severe enough, as observed in patients with CKD or who have received acute high doses of IV magnesium sulfate (used in obstetrics), can activate the CaSR and inhibit PTH secretion.¹⁹²

Hypomagnesemic patients typically have low or inappropriately normal PTH levels for the degree of hypocalcemia observed.¹⁹³ Moderate hypomagnesemia (serum magnesium

levels = 0.8–1 mg/dL) primarily causes PTH resistance at the level of target organ,¹⁹⁴ whereas severe hypomagnesemia, in addition, decreases PTH secretion.¹⁹⁵ The effect of chronic severe hypomagnesemia is not from an extracellular effect on CaSR but from intracellular magnesium depletion, which leads to G_{αs} activation, enhanced CaSR signaling and, hence, blunted PTH secretion.¹⁹⁵ The appropriate therapy is magnesium repletion; in the absence of adequate magnesium repletion, the hypocalcemia is resistant to PTH and vitamin D therapy.

Autoimmune Disease. Autoimmune HP can present as an isolated finding or as part of the polyendocrinopathy type 1 syndrome (APS1). APS1 can be sporadic or familial (also referred to as APECED; see Table 18.2).¹⁹⁶ Autoantibodies against parathyroid tissue have been reported in a significant percentage of cases of hypoparathyroidism, but the causative role of these antibodies is unclear. The CaSR has been identified as a possible autoantigen in some cases of autoimmune HP (isolated or polyglandular).¹⁹⁷

Vitamin D-Related Disorders

Low Vitamin D Production. Inherited and acquired disorders of vitamin D and its metabolites can be associated with hypocalcemia.¹⁹⁸ Vitamin D is a fat-soluble vitamin that is produced in the skin under UVB radiation from 7-dehydrocholesterol or absorbed in the gastrointestinal tract from external sources. Vitamin D is present naturally in a few foods, is artificially added to others, and is available as a food supplement or drug.¹⁹⁹

Despite routine dietary supplementation in milk and other foods, vitamin D deficiency is common in certain populations,^{198,199} such as breastfeeding infants, older adults, people with dark skin and limited sun exposure, people with fat malabsorption,¹⁹⁹ and patients after gastric bypass surgery.²⁰⁰ A study of hospitalized patients has found a high prevalence of vitamin D deficiency, even in younger patients without risk factors who were consuming the recommended daily allowance of vitamin D.²⁰¹ Fat malabsorption syndromes, common in liver diseases, sprue, and Whipple and Crohn diseases, may result in malabsorption of vitamin D.^{202,203} Liver diseases may impair the hydroxylation of vitamin D to 25(OH)D (calcidiol), and drugs such as phenytoin and barbiturates stimulate the conversion of 25(OH)D to inactive metabolites.²⁰⁴ Therapy of hepatic osteodystrophy with vitamin D and calcium is not fully effective.²⁰⁵

Deficiency of 1 α -hydroxylase, as observed in advanced CKD, leads to a deficiency of 1,25(OH)₂D (calcitriol), the most important biologic form for maintaining calcium and phosphorus homeostasis. Vitamin D deficiency with hypocalcemia is commonly seen in patients with kidney disease (see Chapter 55). Patients with nephrotic syndrome may also have decreased 25(OH)D levels as a result of urinary loss, leading to hypocalcemia and secondary HPT.²⁰⁶

The serum level of 25(OH)D is the best indicator of vitamin D status. Levels of 1,25(OH)₂D do not decrease until vitamin D deficiency is severe. Prolonged vitamin D deficiency causes rickets in children (a disorder of mineralization of growing bone) and osteomalacia in adults (a disorder of mineralization of formed bone). The combination of calcium deficiency and vitamin D deficiency accelerates skeletal abnormalities and the development of hypocalcemia. The diagnosis of

vitamin D deficiency is confirmed by measurement of the serum 25(OH)D levels. Hypocalcemia can be observed only in severe vitamin D deficiency—25(OH)D levels less than 10 ng/mL—and when skeletal stores of calcium are depleted; otherwise, the compensatory rise of PTH would be able to mobilize calcium from bone.^{198,199,207,208} The 24-hour urinary calcium excretion is low to very low. Hypophosphatemia and increased alkaline phosphatase and normal FGF-23 levels are typically seen with vitamin D deficiency.¹⁹⁸

Vitamin D Resistance. The observation that some forms of rickets cannot be cured by regular doses of vitamin D has led to the discovery of rare inherited abnormalities in vitamin D metabolism or the vitamin D receptor. Vitamin D–dependent rickets type 1 (VDDR-1; Online Mendelian Inheritance in Man [OMIM database] 264700) is characterized by autosomal recessive, childhood-onset rickets, hypocalcemia, secondary HPT, and aminoaciduria. The biochemical abnormality is defective 1 α -hydroxylation of 25(OH)D, caused by mutations in the gene for the 25(OH)D–1 α -hydroxylase.²⁰⁹ Therapy with calcitriol or 1 α (OH)D (alfacalcidol) restores serum 1,25(OH)₂D and must be continued for life.

VDDR-2 (also called “hereditary vitamin D–resistant rickets”) is an autosomal recessive disorder (OMIM 277440). Affected patients have extreme elevations in 1,25(OH)₂D levels, in addition to alopecia and the abnormalities seen in VDDR-1.²¹⁰ Biochemically, the disorder results from end-organ resistance to 1,25(OH)₂D. A number of different mutations have been found in the vitamin D receptor gene of affected individuals.²¹¹ High-dose calcium intake and calcium infusion may be the only way to treat hypocalcemia and rickets in these children.

Miscellaneous Causes

Medications. Medication-induced hypocalcemia is a relatively common cause of hypocalcemia, particularly in hospitalized patients.^{212,213} Some of the gadolinium-based contrast agents (e.g., gadodiamide, gadoversetamide) used in magnetic resonance imaging (MRI) studies cause pseudohypocalcemia by interference with colorimetric assays for calcium. Calcium readings can be as low as 6 mg/dL, but with no symptoms or signs of hypocalcemia.^{212,213} Propofol and IV contrast agents may also complex calcium.

Drug-induced hypomagnesemia (e.g., cisplatin, aminoglycoside, amphotericin, diuretics) and hypermagnesemia (e.g., magnesium sulfate infusion, magnesium-containing antacids) can cause hypocalcemia. Inhibitors of bone resorption (e.g., bisphosphonates, denosumab, calcimimetics, mithramycin, calcitonin) may depress serum calcium to subnormal levels.²¹² Proton pump inhibitors and histamine-2 antagonists may reduce calcium absorption, provoke hypocalcemia, and/or inhibit bone resorption.²¹⁴ Regional citrate anticoagulation for continuous kidney replacement therapy (CRRT) and for plasmapheresis can chelate calcium and cause hypocalcemia.^{215,216} Transfusions of citrated blood rarely cause significant hypocalcemia, but it may occur in the course of a massive transfusion.²¹⁷

Foscarnet (trisodium phosphonoformate), an antiviral medication used to treat herpes viruses, is a structural mimic of the pyrophosphate anion. Foscarnet can cause hypocalcemia through the chelation of extracellular calcium ions, so normal total calcium measurements may not reflect

ionized hypocalcemia.²¹⁸ Magnesium losses by the kidney may exaggerate hypocalcemia. Patients treated with foscarnet should undergo total calcium and ionized calcium measurements. As noted, anticonvulsants, particularly phenytoin and phenobarbital, can induce the hepatic CYP3A4 enzyme, which shortens vitamin D half-life and causes vitamin D deficiency.²¹³ Fluoride overdose is an exceedingly rare cause of hypocalcemia. Oral sodium phosphate–induced hyperphosphatemia may cause hypocalcemia, particularly in patients with CKD.^{212,219} Other drugs associated with hypocalcemia include antiinfectious (e.g., pentamidine, ketoconazole) and chemotherapeutic agents (e.g., asparaginase, cisplatin, doxorubicin).

Critical Illness. In complicated, critically ill patients, total calcium measurements may be poor indicators of the ionized calcium concentration because many factors that could interfere with or alter calcium and protein binding may be present (e.g., albumin infusion, citrate, IV fluids, acid-base disturbances, dialysis therapy, propofol infusion). Thus, it is particularly important to measure ionized calcium in this setting. Hypocalcemia is frequently noted in gram-negative sepsis and toxic shock syndrome.^{150,220} This entity is multifactorial; the primary cause is unclear, but a direct effect of IL-1 on parathyroid function may be partly responsible.²²¹

Other Causes. Hypocalcemia is common in acute pancreatitis and is a poor prognostic indicator. It is probably due to calcium chelation by free fatty acids generated by the action of pancreatic lipase, although some animal studies have challenged this hypothesis.²²² Massive tumor lysis, particularly from rapidly growing hematologic malignancies, may cause hyperphosphatemia, hyperuricemia, and hypocalcemia.²²³ The early phase of rhabdomyolysis may include severe hyperphosphatemia and associated hypocalcemia, in contrast to the recovery phase, when hypercalcemia is common. In hemodialysis patients, hypocalcemia is common and may result, at least in part, from reduced kidney phosphate clearance and consequent hyperphosphatemia and reduced 1,25(OH)₂D production (see Chapter 55).

MANAGEMENT OF HYPOCALCEMIA

The optimal management of hypocalcemia has not been examined in clinical trials, but there are accepted practices. The treatment depends on the speed of onset and severity of clinical and laboratory features. Oral calcium supplementation may be sufficient treatment for mild hypocalcemia. Patients with acute, severe symptomatic hypocalcemia (Ca level <7–7.5 mg/dL; ionized Ca²⁺ < 0.8 mmol/L), such as after parathyroidectomy, with evidence of neuromuscular effects or tetany, should be treated promptly with IV calcium. The preferred calcium salt is calcium gluconate (10 mL of 10% calcium gluconate contains 93 mg of elemental calcium). Initially, 1 to 2 g (93–186 mg of elemental Ca) of IV calcium gluconate in 50 mL of 5% dextrose is given over a period of 10 to 20 minutes, followed by slow infusion at a rate of 0.3 to 1.0 mg elemental Ca/kg per hour.¹⁹⁴ The dose can be adjusted to maintain the serum calcium level at the lower end of normal values.

Moderate asymptomatic hypocalcemia (ionized Ca²⁺ >0.8 mmol/L) can be treated by repeated doses of 1 to 2 g calcium gluconate IV every 4 hours, without continuous

infusion. Ionized Ca should be measured every 4 to 6 hours initially.

The infusion solution should not contain phosphates or bicarbonates. IV infusion may be needed until the patient is stabilized, and oral calcium and calcitriol therapy begin to take effect. Correction of hypomagnesemia and hyperphosphatemia should also be undertaken, when present. Dialysis may be appropriate if hyperphosphatemia is also present.

Treatment of chronic hypocalcemia depends on the underlying cause. For example, underlying hypomagnesemia or vitamin D deficiency should be corrected. The principal therapy for primary parathyroid dysfunction or PTH resistance is dietary calcium supplementation and vitamin D therapy. Oral calcium supplementation, beginning with 500 to 1000 mg of elemental calcium daily and increasing up to a maximum of 2000 mg daily, is a good strategy. Correction of the serum calcium level to the low-normal range is generally advised; correction to normal levels may lead to frank hypercalciuria. Several preparations of vitamin D are available for the treatment of hypocalcemia. The role of vitamin D therapy in CKD is discussed separately.

Replacement therapy using synthetic human parathyroid hormone, PTH (1-34) (teriparatide, 20 µg subcutaneously once daily) has been FDA-approved for the treatment of osteoporosis. Teriparatide has also been used as hormone replacement therapy in patients with hypoparathyroidism in a dose of 20 µg subcutaneously, twice daily.²²⁴ More recently, PTH (1-84) (100 µg every other day) was studied in 30 hypoparathyroid patients for 24 months.²²⁵ Improvement or normalization of the serum calcium level was observed with both hormone preparations.

DISORDERS OF MAGNESIUM HOMEOSTASIS

HYPOMAGNESEMIA AND MAGNESIUM DEFICIENCY

The terms “hypomagnesemia” and “magnesium deficiency” tend to be used interchangeably. In marked contrast with the distribution of calcium, extracellular fluid magnesium accounts for only 1% of total body magnesium, so serum magnesium concentrations have been found to correlate poorly with overall magnesium status. In patients with magnesium deficiency, serum magnesium concentrations may be normal or may seriously underestimate the severity of the magnesium deficit.²²⁶ Approximately 50% to 60% of magnesium is in the skeleton, and most of the remaining 40% to 50% is intracellular. No satisfactory clinical test to assay body magnesium stores is available.²²⁷

The magnesium tolerance test has been proposed to be the best test of overall magnesium status.²²⁷ It is based on the observation that magnesium-deficient patients tend to retain a greater proportion of a parenterally administered magnesium load and excrete less in the urine than normal individuals.²²⁸ Clinical studies have indicated that the results of a magnesium tolerance test correlate well with magnesium status, as assessed by skeletal muscle magnesium content and exchangeable magnesium pools. However, the test is invalid in patients who have impaired kidney function or a kidney

magnesium-wasting syndrome or in patients taking diuretics or other medications that induce kidney magnesium wasting. Thus, and also because of the time and effort required to perform the magnesium tolerance test, it is used primarily as a research tool.

The serum magnesium concentration, although an insensitive measure of magnesium deficit, remains the only practical test of magnesium status in widespread use. Surveys of serum magnesium levels in hospitalized patients have indicated a high incidence of hypomagnesemia (presumably an underestimate of the true incidence of magnesium deficiency), ranging from 11% in general inpatients²²⁹ to 60% in patients admitted to intensive care units (ICUs).^{230,231} Furthermore, in ICU patients, hypomagnesemia was associated with increased mortality when compared with normomagnesemic patients.²³⁰

The functionally important value is believed to be the ionized Mg^{2+} concentration, which is less than total serum magnesium due to protein binding. Measurements with ion-selective electrodes have found ionized Mg^{2+} concentrations that are approximately 70% of the total serum magnesium, a proportion that is fairly constant in the general population.²²⁷ However, in critically ill patients, there is a poor correlation between total and ionized serum magnesium levels.²³¹ It is surprising that we rarely measure ionized serum magnesium.

CAUSES

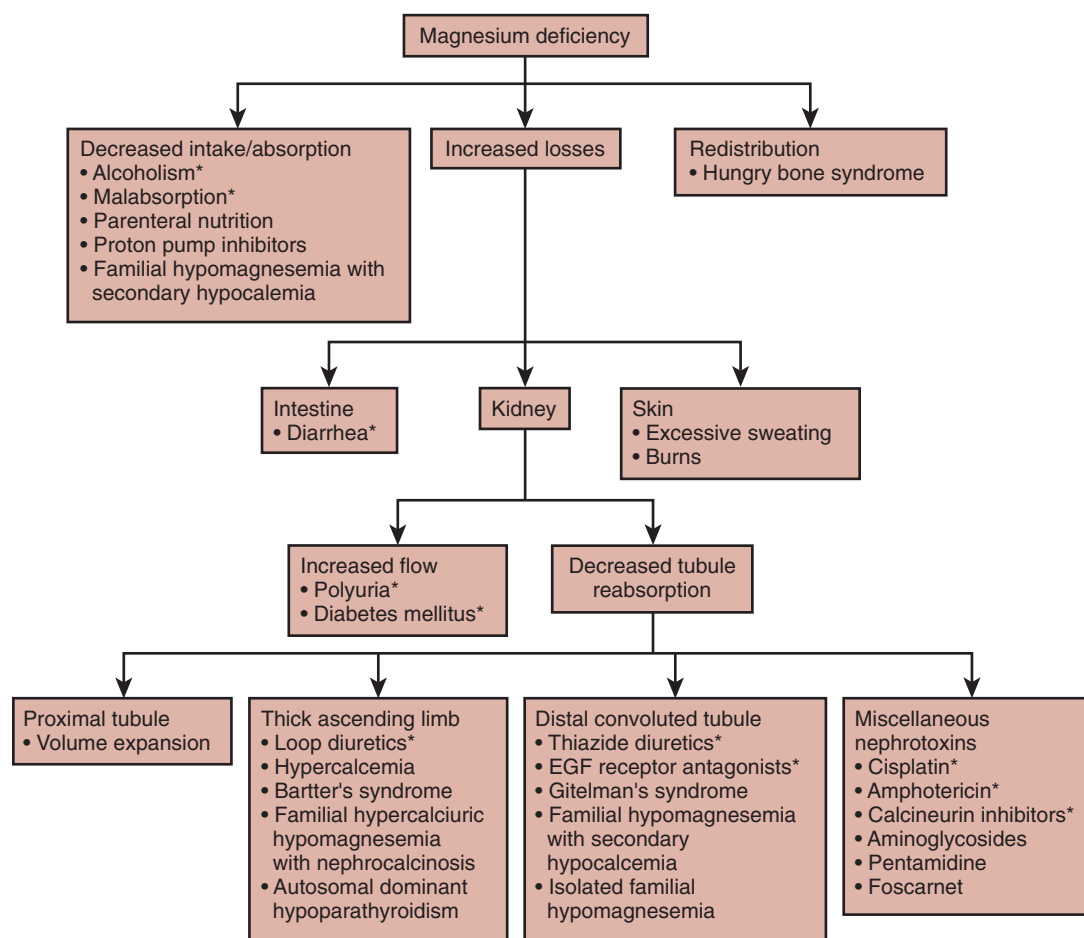
Magnesium deficiency may be caused by decreased intake or intestinal absorption, increased losses via the gastrointestinal tract, kidneys, or skin or, rarely, sequestration in the bone compartment (Fig. 18.4). It is often helpful to distinguish between kidney magnesium wasting and extrarenal causes of magnesium deficiency by assessing urinary magnesium excretion. In the setting of magnesium deficiency, a urine magnesium excretion rate greater than 24 mg/day is abnormal and is usually suggestive of kidney magnesium wasting.²³² If a 24-hour urine collection is unavailable, the fractional excretion of magnesium (F_{EMg}) can be calculated from a random urine specimen as follows:

$$F_{EMg} = \frac{\text{urine Mg concentration} \times \text{plasma Cr}}{([0.7 \times \text{plasma total Mg concentration}] \times \text{urine Cr concentration})}$$

where Cr is creatinine. Note that a correction factor of 0.7 is applied to the plasma total magnesium concentration to estimate the free magnesium concentration. In general, a F_{EMg} value of more than 3% to 4% in an individual with a normal GFR is indicative of inappropriate urinary magnesium loss.¹⁹³ If kidney magnesium wasting has been excluded, the losses must be extrarenal in origin, and the underlying cause can usually be identified from the case history.

Extrarenal Causes

Nutritional Deficiency. Development of magnesium deficiency due to dietary deficiency in normal individuals is unusual because almost all foods contain significant amounts of magnesium, and kidney adaptation to conserve magnesium is very efficient. Thus, magnesium deficiency of nutritional origin is observed primarily in two clinical settings—alcoholism and parenteral feeding.



*Common causes.

Fig. 18.4 Causes of magnesium deficiency. EGF, Epidermal growth factor.

In chronic alcoholics, the intake of ethanol substitutes for the intake of important nutrients.²³³ Approximately 20% to 25% of alcoholics are frankly hypomagnesemic, and most can be shown to be magnesium-deficient with the magnesium tolerance test.²²⁸ Alcohol also impairs kidney magnesium reabsorption.²³⁴

Patients receiving parenteral nutrition may also develop hypomagnesemia.²³⁵ In general, these patients are sicker than the average inpatient and are more likely to have other conditions associated with a magnesium deficit and ongoing magnesium losses.

Hypomagnesemia may also be a consequence of the refeeding syndrome.²³⁶ In this condition, overzealous parenteral feeding of severely malnourished patients causes hyperinsulinemia, as well as a rapid cellular uptake of glucose and water, together with phosphorus, potassium, and magnesium.

Intestinal Malabsorption. Generalized malabsorption syndromes caused by conditions such as celiac disease, Whipple disease, and inflammatory bowel disease are frequently associated with intestinal magnesium wasting and magnesium deficiency.²³⁷ In fat malabsorption with concomitant steatorrhea, free fatty acids in the intestinal lumen may combine with magnesium to form nonabsorbable soaps, a process

termed “saponification,” thus contributing to impaired magnesium absorption. The severity of hypomagnesemia in patients with malabsorption syndrome correlates with the fecal fat excretion rate and, in rare patients, reduction of dietary fat intake alone, which reduces steatorrhea, can correct the hypomagnesemia. Previous intestinal resection, particularly of the distal part of the small intestine, is also an important cause of magnesium malabsorption.²³⁸ Magnesium deficiency was a common complication of bariatric surgery by jejunoileal bypass,²³⁹ but fortunately does not occur with the modern technique of gastric bypass.²⁴⁰

Proton pump inhibitors (PPIs) have been reported to cause hypomagnesemia due to intestinal magnesium malabsorption.²⁴¹ Among patients admitted to an ICU, concurrent use of PPIs with diuretics has been associated with a significant increase of hypomagnesemia (odds ratio [OR], 1.54) and a 0.03-mg/dL lower serum magnesium concentration compared with patients taking diuretics alone, whereas patients taking PPI alone did not have an increased risk of hypomagnesemia.²⁴² Interestingly, in a recent case-control study of hospitalized patients, hypomagnesemia at the time of hospital admission was not associated with out of hospital use of PPI.²⁴³ However, most of these patients were not taking diuretics. A rare mutation of the TRPM6 magnesium transport channel can also lead to intestinal magnesium malabsorption, along

with kidney magnesium wasting, causing hypomagnesemia with secondary hypocalcemia.²⁴⁴

Diarrhea and Gastrointestinal Fistula. The magnesium concentration of diarrheal fluid is high and ranges from 1 to 16 mg/dL,²³⁸ so magnesium deficiency may occur in patients with chronic diarrhea of any cause, even in the absence of concomitant malabsorption.²²⁶ It also occurs in patients who abuse laxatives. By contrast, secretions from the upper gastrointestinal tract are low in magnesium content, and significant magnesium deficiency is, therefore, rarely observed in patients with an intestinal, biliary, or pancreatic fistula, ileostomy, or prolonged gastric drainage (except as a consequence of malnutrition).²³⁸

Cutaneous Losses. Hypomagnesemia may be observed after prolonged intense exertion. For example, serum magnesium concentrations fall 20% on average after a marathon run.²⁴⁵ About 25% of the decrease in the serum magnesium level can be accounted for by losses in sweat, which can contain up to 0.5 mg/dL of magnesium; the remainder is most likely due to transient redistribution into the intracellular space. Magnesium supplements may be indicated in a number of sports, especially if the athlete is on a suboptimal magnesium diet.²⁴⁶

Hypomagnesemia occurs in 40% of patients with severe burn injuries during the early period of recovery. The major cause is loss of magnesium in the cutaneous exudate, which can exceed 1 g/day.²⁴⁷

Redistribution to Bone Compartment. Hypomagnesemia may occasionally accompany the profound hypocalcemia of hungry bone syndrome observed in some patients with HPT and severe bone disease immediately after parathyroidectomy.²⁴⁸ In such cases, there is high bone turnover, and sudden removal of excess PTH is believed to result in virtual cessation of bone resorption, with a continued high rate of bone formation and consequent sequestration of calcium and magnesium into bone mineral.

Diabetes Mellitus. Hypomagnesemia is common in patients with diabetes mellitus and has been reported to occur in 13.5% to 47.7% of nonhospitalized patients with type 2 diabetes.²⁴⁹ The cause is thought to be multifactorial; contributing factors include decreased oral intake of magnesium-rich foods, poor intestinal absorption due to diabetic autonomic neuropathy, and increased kidney excretion. The latter could, in turn, be caused by glomerular hyperfiltration, osmotic diuresis, or decreased TAL and distal tubule magnesium reabsorption caused by functional insulin deficiency.^{250,251} In addition, some studies have suggested that magnesium deficiency might itself impair glucose tolerance, thus partly explaining the association. Conversely, genetic variants in the magnesium transport channels, TRPM6 and TRPM7, may increase the risk of type 2 diabetes mellitus in women on a diet with less than 250 mg/day of magnesium.²⁵²

Renal Magnesium Wasting

The diagnosis of kidney magnesium wasting is made by demonstrating an inappropriately high rate of kidney magnesium excretion in the face of hypomagnesemia, as described earlier. The causes are summarized in Fig. 18.4.

Polyuria. Increased urine output from any cause is often accompanied by increased kidney losses of magnesium. Kidney magnesium wasting occurs with osmotic diuresis—for example, in hyperglycemic crises in diabetics.^{253,254} Hypermagnesuria also occurs during the polyuric phase of recovery from acute kidney injury (AKI) in a native kidney, during recovery from ischemic injury in a transplanted kidney, and in postobstructive diuresis. In such cases, it is likely that residual tubule reabsorptive defects persisting from the primary kidney injury play as important a role as polyuria itself in inducing kidney magnesium wasting.²⁵⁵

Extracellular Fluid Volume Expansion. In the proximal tubule, magnesium reabsorption is passive and driven by the reabsorption of sodium and water in this segment. Extracellular volume expansion, which decreases proximal sodium and water reabsorption, also increases urinary magnesium excretion. Thus chronic therapy with magnesium-free parenteral fluids, crystalloid or hyperalimentation,²⁵⁶ can cause kidney magnesium wasting, as can hyperaldosteronism.²⁵⁷

Diuretics. Loop diuretics inhibit the apical membrane $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter of the TAL and abolish the trans-epithelial potential difference, thereby inhibiting paracellular magnesium reabsorption. Hypomagnesemia is, therefore, a frequent finding in patients undergoing chronic loop diuretic therapy.²⁵⁸ Chronic treatment with thiazide diuretics, which inhibit the NaCl cotransporter (NCC), also causes kidney magnesium wasting. Thiazide diuretics or knockout of NCC in mice causes downregulation of expression of the apical magnesium entry channel in the distal convoluted tubule, TRPM6, which may explain the mechanism of the magnesuria.²⁵⁹

Epidermal Growth Factor Receptor Blockers. Hypomagnesemia is common in patients receiving cetuximab²⁶⁰ and panitumumab,²⁶¹ which are monoclonal blocking antibodies of the epidermal growth factor (EGF) receptor used in the treatment of metastatic colorectal cancer. The incidence of hypomagnesemia increases with increasing duration of therapy, reaching almost 50% in patients treated for longer than 6 months.²⁶² The median time to onset of hypomagnesemia after beginning treatment is 99 days, and it generally reverses 1 to 3 months after discontinuing therapy.²⁶³ F_eMg is inappropriately elevated, suggesting a defect in kidney magnesium reabsorption.²⁶⁴ Studies have suggested that the EGF receptor is located basolaterally in the distal convoluted tubule (DCT).²⁶⁴ Autocrine or paracrine activation of the receptor stimulates redistribution of TRPM6 to the apical membrane via a Rac1 -dependent signaling pathway²⁶⁵ and presumably increases transepithelial magnesium reabsorption. Thus EGF receptor blockade likely causes kidney magnesium wasting by antagonizing this pathway.

Hypercalcemia. Elevated serum ionized Ca^{2+} levels—for example, in patients with malignant bone metastases—directly induce kidney magnesium wasting and hypomagnesemia,²⁶⁶ probably by stimulating the basolateral CaSR in the TAL of Henle. In HPT, the situation is more complicated because the hypercalcemia-induced tendency to Mg wasting is counteracted by the action of PTH to stimulate magnesium reabsorption; thus, kidney magnesium handling is usually normal, and magnesium deficiency is rare.²⁶⁷

Tubule Nephrotoxins. Cisplatin, a widely used chemotherapeutic agent for solid tumors, frequently causes kidney magnesium wasting. Hypomagnesemia is almost universal at a monthly dose of 50 mg/m².²⁶⁸ The occurrence of Mg wasting does not appear to correlate with the incidence of cisplatin-induced AKI.²⁶⁹ Kidney magnesuria continues after cessation of the drug for a mean of 4 to 5 months, but can persist for years.²⁶⁹ Although the nephrotoxic effects of cisplatin are manifested histologically as acute tubular necrosis confined to the S3 segment of the proximal tubule, the magnesuria does not correlate temporally with the clinical development of AKI secondary to acute tubular necrosis. Furthermore, patients who become hypomagnesemic are also subject to the development of hypocalciuria, thus suggesting that the reabsorption defect may actually be in the DCT. Mouse studies have also suggested that cisplatin may reduce the expression of transport proteins in the DCT.²⁷⁰ Cisplatin may also impair intestinal magnesium absorption.²⁷¹ Carboplatin, an analogue of cisplatin, is considerably less nephrotoxic, and only rarely causes AKI or hypomagnesemia.²⁷²

Amphotericin B is a well-recognized tubule nephrotoxin that can cause kidney potassium wasting, distal kidney tubular acidosis, and AKI, with tubule necrosis and calcium deposition noted in the DCT and TAL on kidney biopsy.²⁷³ Amphotericin B causes kidney magnesium wasting and hypomagnesemia related to the cumulative dose administered, but these effects may be observed after as little as a 200-mg total dose.²⁷⁴ Interestingly, the amphotericin-induced magnesuria is accompanied by the reciprocal development of hypocalciuria so, as with cisplatin, the serum calcium concentration is usually preserved, again suggesting that the functional tubule defect resides in the DCT.

Aminoglycosides cause a syndrome of kidney magnesium and potassium wasting with hypomagnesemia, hypokalemia, hypocalcemia, and tetany. Hypomagnesemia may occur, despite levels in the appropriate therapeutic range.²⁷⁵ Most patients have reported that they had delayed onset of hypomagnesemia occurring after at least 2 weeks of therapy, and received total doses in excess of 8 g, thus suggesting that it is the cumulative dose of aminoglycoside that is the key predictor of toxicity. In addition, no correlation was found between the occurrence of aminoglycoside-induced acute tubular necrosis and hypomagnesemia. Magnesium wasting persists after cessation of the aminoglycoside, often for several months. All aminoglycosides in clinical use have been implicated, including gentamicin, tobramycin, and amikacin, as well as neomycin when administered topically for extensive burn injuries. This form of symptomatic aminoglycoside-induced kidney magnesium wasting is now relatively uncommon because of heightened general awareness of its toxicity. However, asymptomatic hypomagnesemia can be observed in one-third of those treated with a single course of an aminoglycoside at standard doses (3–5 mg/kg/day, for a mean of 10 days). In these cases, the hypomagnesemia occurs on average 3 to 4 days after the start of therapy and readily reverses after cessation of therapy.²⁷⁶

IV pentamidine causes hypomagnesemia as a result of kidney magnesium wasting in most patients, typically in association with hypocalcemia.²⁷⁷ The average onset of symptomatic hypomagnesemia occurs after 9 days of therapy, and the defect persists for at least 1 to 2 months after the discontinuation of pentamidine. Hypomagnesemia is also

observed in two-thirds of AIDS patients with cytomegalovirus retinitis treated intravenously with the pyrophosphate analogue foscarnet.²⁷⁸ As with aminoglycosides and pentamidine, foscarnet-induced hypomagnesemia is often associated with significant hypocalcemia.

The calcineurin inhibitors cyclosporine and tacrolimus cause kidney magnesium wasting and hypomagnesemia in patients after organ transplantation.²⁷⁹ The mechanism is thought to be downregulation of the distal tubule magnesium channel, TRPM6.²⁸⁰

Tubulointerstitial Nephropathies. Kidney Mg wasting has occasionally been reported in patients with acute or chronic tubulointerstitial nephritis that is not caused by nephrotoxic drugs—for example, in chronic pyelonephritis and acute kidney allograft rejection. Other manifestations of tubule dysfunction, such as salt wasting, hypokalemia, kidney tubular acidosis, and Fanconi syndrome, may also be present and provide clues to the diagnosis.²⁵⁵

Inherited Kidney Magnesium-Wasting Disorders

Primary Magnesium-Wasting Disorders. Primary magnesium-wasting disorders are rare. Patients can be broadly classified into distinct clinical syndromes, depending on whether the hypomagnesemia is isolated, occurs together with hypocalcemia, or is associated with hypercalciuria and nephrocalcinosis.²⁸¹ The pathogenesis and clinical features of these syndromes, which generally present in childhood, are discussed in detail in Chapter 73.

Bartter and Gitelman Syndromes. Bartter syndrome is an autosomal recessive disorder characterized by Na wasting, hypokalemic metabolic alkalosis, and hypercalciuria that usually occurs in infancy or early childhood (see also Chapter 44).²⁸² All Bartter syndrome patients are by definition hypercalciuric and, in addition, one-third have hypomagnesemia with inappropriate magnesuria, consistent with loss of the TAL transepithelial potential difference that drives paracellular divalent cation reabsorption. Thus the physiology of Bartter syndrome is essentially identical to that of chronic loop diuretic therapy. Gitelman syndrome is a variant of Bartter syndrome distinguished primarily by hypocalciuria.²⁸³ Patients with Gitelman syndrome are identified later in life, usually after the age of 6 years, and have milder symptoms. The genetic defect in these families is caused by inactivating mutations in the DCT electroneutral thiazide-sensitive NCC and therefore resembles chronic thiazide diuretic therapy. Kidney magnesium wasting and hypomagnesemia are universally found in patients with Gitelman syndrome.

Calcium-Sensing Disorders. In FHH, the hypercalcemia is due to inactivating mutations in CaSR (discussed earlier). As a consequence of the inactivated CaSR, the normal magnesuric response to hypercalcemia is impaired,²⁸⁴ so these patients are paradoxically mildly hypermagnesemic. Activating mutations in CaSR cause the opposite syndrome, autosomal dominant hypoparathyroidism. As might be expected, most of these patients are mildly hypomagnesemic, presumably because of TAL magnesium wasting.¹⁶⁸

CLINICAL MANIFESTATIONS

Hypomagnesemia may cause symptoms and signs of disordered cardiac, neuromuscular, and central nervous system function. It is also associated with an imbalance of other electrolytes,

such as potassium and calcium. Many of the cardiac and neurologic manifestations attributed to magnesium deficiency may be explained by the frequent coexistence of hypokalemia and hypocalcemia in the same patient. Patients with mild hypomagnesemia or who are magnesium-deficient with normal serum magnesium levels may be completely asymptomatic.²⁸⁵ Thus, the clinical importance of mild to moderate magnesium depletion remains controversial, although it has been associated with a number of disorders, such as hypertension and osteoporosis (see later).

Cardiovascular System

Magnesium has protean and complex effects on myocardial ion fluxes. Because magnesium is an obligate cofactor in all reactions that require ATP, it is essential for the activity of $\text{Na}^+\text{-K}^+\text{-ATPase}$.²⁸⁶ During magnesium deficiency, $\text{Na}^+\text{-K}^+\text{-ATPase}$ function is impaired. The intracellular potassium concentration falls, which may potentially result in a relatively depolarized resting membrane potential and predispose to ectopic excitation and tachyarrhythmias.²⁸⁷ Furthermore, the magnitude of the outward potassium gradient is decreased, thereby reducing the driving force for the potassium efflux needed to terminate the cardiac action potential and, as a result, repolarization is delayed. Changes in the ECG may be observed with isolated hypomagnesemia and usually reflect abnormal cardiac repolarization, including bifid T waves and other nonspecific abnormalities in T wave morphology, U waves, prolongation of the QT or QU interval and, rarely, electrical alternation of the T or U wave.²⁸⁸

Numerous anecdotal reports have indicated that hypomagnesemia alone can predispose to cardiac tachyarrhythmias, particularly of ventricular origin, including torsades de pointes, monomorphic ventricular tachycardia, and ventricular fibrillation, which may be resistant to standard therapy and respond only to magnesium repletion.²⁸⁸ Many of the reported patients also had a prolonged QT interval, an abnormality known to predispose to torsades de pointes, and may also increase the period of vulnerability to the R-on-T phenomenon. In the setting of exaggerated cardiac excitability, hypomagnesemia may be the trigger for other types of ventricular tachyarrhythmias.²⁸⁸ In addition, hypomagnesemia, like hypokalemia, facilitates the development of digoxin cardiotoxicity.²⁸⁹ Because cardiac glycosides and magnesium depletion inhibit $\text{Na}^+\text{-K}^+\text{-ATPase}$, their additive effects on intracellular potassium depletion may account for their enhanced toxicity in combination.

It is clear that patients with underlying cardiac disease who have severe hypomagnesemia, particularly in combination with hypokalemia, may develop arrhythmias. The issue of whether mild isolated hypomagnesemia and magnesium depletion in individuals without overt heart disease carries the same risk has been controversial.²⁹⁰ In one small prospective study, low dietary magnesium appeared to increase the risk for supraventricular and ventricular ectopy, despite the absence of frank hypomagnesemia, hypokalemia, and hypocalcemia.²⁹¹ In the Framingham Offspring Study, lower levels of serum magnesium were associated with a higher prevalence of ventricular premature complexes.²⁹² A low serum magnesium level is also associated with the development of atrial fibrillation.²⁹³ In the Framingham Offspring Study, individuals in the lowest quartile of serum magnesium were, after up to 20 years of follow-up, approximately 50% more likely to

develop atrial fibrillation compared with those in the upper quartiles.

Several large population-based studies have shown a strong association between low serum magnesium levels and increased cardiovascular and all-cause mortality.^{294–296} Higher magnesium intake is associated with reduced risk of coronary heart disease and cardiac death, stroke, and coronary calcification.^{297–302}

An inverse relationship between dietary magnesium intake and blood pressure has also been observed.^{303–306} Hypomagnesemia and/or reduction of intracellular magnesium have also been inversely correlated with blood pressure. This may be especially important in diabetes mellitus.³⁰⁷ Patients with essential hypertension were found to have reduced free magnesium concentrations in red blood cells. The magnesium levels were inversely related to systolic and diastolic blood pressures. Intervention studies with magnesium therapy in hypertension have led to conflicting results. Several have shown a positive blood pressure-lowering effect of supplements, but others have not. Other dietary factors may also play a role. In the DASH study, a diet rich in fruits and vegetables, which increased magnesium intake from 176 to 423 mg/day (along with an increase in potassium), significantly lowered blood pressure.³⁰⁸ The mechanism whereby magnesium deficit may affect blood pressure is not clear, but magnesium does regulate vascular tone and reactivity and attenuates agonist-induced vasoconstriction. Magnesium depletion may involve decreased production of prostacyclin, increased production of thromboxane A₂, and enhanced vasoconstrictive effects of angiotensin II and norepinephrine. Importantly, in none of these studies has magnesium therapy been rigorously studied, whether as a therapy for blood pressure reduction or prophylaxis against cardiovascular disease, arrhythmias, or stroke.

The only setting in which the role of magnesium deficiency and clinical utility of adjunctive magnesium therapy has been studied extensively is in acute myocardial infarction (AMI). Magnesium deficiency may be a risk factor because it has been shown to play a role in systemic and coronary vascular tone, in cardiac dysrhythmias (see earlier), and in the inhibition of steps in the coagulation process and platelet aggregation. Although several small controlled trials have suggested that adjunctive magnesium therapy reduces mortality from AMI by 50%, three major trials have defined our understanding of magnesium therapy in AMI.³⁰⁹ The LIMIT-2 study was the first study involving large numbers of participants. Magnesium treatment showed an approximately 25% lower mortality rate. In the Fourth International Study of Infarct, unlike LIMIT-2, the mortality rate in the magnesium-treated group was not significantly different from that in the control group. The most recently published Magnesium in Coronaries (MAGIC) trial was designed to address early intervention in higher risk patients.³⁰⁹ Over a 3-year period, 6213 participants were studied. The magnesium-treated group mortality at 30 days was not significantly different from that of those given placebo. The overall evidence from clinical trials does not support the routine application of adjunctive magnesium therapy in patients with AMI at this time.

Neuromuscular System

Symptoms and signs of neuromuscular irritability, including tremor, muscle twitching, Trousseau and Chvostek signs, and frank tetany, may develop in patients with isolated

hypomagnesemia³¹⁰ and in patients with concomitant hypocalcemia. Hypomagnesemia is also frequently manifested as seizures, which may be generalized and tonic-clonic in nature or may be multifocal motor seizures, and they are sometimes triggered by loud noises.³¹⁰ Interestingly, noise-induced seizures and sudden death are also characteristic of mice made hypomagnesemic by dietary magnesium deprivation. The effects of magnesium deficiency on brain neuronal excitability are thought to be mediated by N-methyl-D-aspartate (NMDA)-type glutamate receptors.³¹¹ Glutamate is the principal excitatory neurotransmitter in the brain; it acts as an agonist at NMDA receptors and opens a cation conductance channel that depolarizes the postsynaptic membrane. Extracellular magnesium normally blocks NMDA receptors, so hypomagnesemia may release the inhibition of glutamate-activated depolarization of the postsynaptic membrane and thereby trigger epileptiform electrical activity.³¹² Vertical nystagmus is a rare but diagnostically useful neurologic sign of severe hypomagnesemia.³¹³ In the absence of a structural lesion of the cerebellar or vestibular pathways, the only recognized metabolic causes are Wernicke encephalopathy and severe magnesium deficiency.³¹³

Skeletal System

Dietary magnesium depletion in animals has been shown to lead to a decrease in skeletal growth and increased skeletal fragility.³¹⁴ A decrease in osteoblastic bone formation and an increase in osteoclastic bone resorption are implicated as the cause of decreased bone mass. In humans, epidemiologic studies have suggested a correlation between bone mass and dietary magnesium intake.³¹⁵ Few studies have been conducted assessing magnesium status in patients with osteoporosis. Low serum and red blood cell (RBC) magnesium concentrations, as well as high retention of parenterally administered magnesium, have suggested a deficit. Low skeletal magnesium content has been observed in some, but not all, studies. The effect of supplements on bone mass has generally led to an increase in bone mineral density, although study design limits useful information. Larger long-term, placebo-controlled, double-blind investigations are required.

There are several potential mechanisms that may account for a decrease in bone mass in magnesium deficiency. Magnesium is mitogenic for bone cell growth; therefore, deficiency may directly result in a decrease in bone formation. It also affects crystal formation—a lack results in a larger, more perfect crystal, which may affect bone strength. Magnesium deficiency may result in a fall in serum PTH and 1,25 (OH)₂ vitamin D levels (see earlier). Because both hormones are trophic for bone, impaired secretion or skeletal resistance may result in osteoporosis. A low serum 1,25 (OH)₂ vitamin D level may also result in decreased intestinal calcium absorption. An observed increased release of inflammatory cytokines in bone may result in the activation of osteoclasts and increased bone resorption in rodents.^{314,316}

Electrolyte Homeostasis

Patients with hypomagnesemia are frequently also hypokalemic. Many of the conditions associated with hypomagnesemia that have been outlined earlier can cause simultaneous magnesium and potassium loss. However, hypomagnesemia by itself can induce hypokalemia in humans and experimental animals, and such patients are often refractory to potassium

repletion until their magnesium deficit is corrected.³¹⁷ The cause of the hypokalemia appears to be increased secretion in the distal nephron.³¹⁸ The mechanism has been attributed to cytosolic magnesium depletion, which would release intracellular block of the apical secretory potassium channel, ROMK.³¹⁹

Hypocalcemia is present in approximately 50% of patients with hypomagnesemia.²⁸⁵ The major cause is impairment of PTH secretion by magnesium deficiency, which is reversed within 24 hours by magnesium repletion.¹⁹⁴ In addition, hypomagnesemic patients also have low circulating 1,25(OH)₂D levels and end-organ resistance to PTH and vitamin D.¹⁹⁴

Other Disorders

Magnesium depletion has also been associated with several other disorders, such as insulin resistance and the metabolic syndrome in type 2 diabetes mellitus.^{307,320} Magnesium deficiency has been associated with migraine headache, and magnesium therapy has been reported to be effective in the treatment of migraine.³²¹ Because magnesium deficiency results in smooth muscle spasm, it has also been implicated in asthma, and magnesium therapy has been effective in asthma in some studies.^{322,323} Finally, a high dietary magnesium intake has been associated with a reduced risk of colon cancer.^{324,325}

TREATMENT

Magnesium deficiency can sometimes be prevented. Individuals whose dietary intake has been reduced or who are being maintained by parenteral nutrition should receive magnesium supplementation. The recommended daily allowance of magnesium for adults is 420 mg (35 mEq) for men and 320 mg (27 mEq) for women.³²⁶ Thus, in the absence of dietary magnesium intake, an appropriate supplement would therefore be one 140-mg tablet of magnesium oxide four to five times daily or the equivalent dose of an alternative oral magnesium-containing salt. Because the oral bioavailability of magnesium is approximately 33% in patients with normal intestinal function, the equivalent parenteral maintenance requirement of magnesium would be 10 mEq daily.

Once symptomatic magnesium deficiency develops, patients should clearly be repleted with magnesium. However, the importance of treating asymptomatic magnesium deficiency remains controversial. Given the clinical manifestations outlined earlier, it seems prudent to replete all magnesium-deficient patients with a significant underlying cardiac or seizure disorder, patients with concurrent severe hypocalcemia or hypokalemia, and patients with isolated asymptomatic hypomagnesemia, if it is severe (<1.4 mg/dL).

Intravenous Replacement

In the inpatient setting, the IV route of administration of magnesium is favored because it is highly effective and inexpensive, and is usually well tolerated. The standard preparation is MgSO₄ • 7H₂O. The initial rate of repletion depends on the urgency of the clinical situation. In a patient who is actively seizing or who has a cardiac arrhythmia, 8 to 16 mEq (1–2 g) may be administered IV over a 2- to 4-minute period; otherwise, a slower rate of repletion is safer. Because the added extracellular magnesium equilibrates slowly with the intracellular compartment, and because kidney excretion of extracellular magnesium exhibits a threshold effect,

approximately 50% of parenterally administered magnesium is excreted in urine.³²⁷

A slower rate and prolonged course of repletion would be expected to decrease these urinary losses and therefore be much more efficient and effective at replenishing body magnesium stores. The magnitude of the magnesium deficit is difficult to gauge clinically and cannot be readily deduced from the serum magnesium concentration. In general, however, the average deficit can be assumed to be 1 to 2 mEq/kg body weight.³²⁷ A simple regimen for nonemergency magnesium repletion is to administer 64 mEq (8 g) of MgSO_4 over the first 24 hours and then 32 mEq (4 g) daily for the next 2 to 6 days. It is important to remember that serum magnesium levels rise early, whereas intracellular stores take longer to replete, so magnesium repletion should continue for at least 1 to 2 days after the serum magnesium level normalizes. In patients with kidney magnesium wasting, additional magnesium may be needed to replace ongoing losses. In patients with a reduced GFR, the rate of repletion should be reduced by 25% to 50%,³²⁷ the patient should be carefully monitored for signs of hypermagnesemia, and the serum magnesium level should be checked frequently.

The main adverse effects of magnesium repletion are due to hypermagnesemia as a consequence of an excessive rate or amount of magnesium administered. These effects include facial flushing, loss of deep tendon reflexes, hypotension, and atrioventricular block. Monitoring the tendon reflexes is a useful bedside test to detect magnesium overdose. In addition, IV administration of large amounts of MgSO_4 results in an acute decrease in the serum ionized Ca^{2+} level,³²⁸ which is related to increased urinary calcium excretion and complexing of calcium by sulfate. Thus, in an asymptomatic patient who is already hypocalcemic, administration of MgSO_4 may further lower the ionized Ca^{2+} level and thereby precipitate tetany.³²⁹

Oral Replacement

Oral magnesium administration is used initially for the repletion of mild cases of hypomagnesemia or for continued replacement of ongoing losses in the outpatient setting after an initial course of IV repletion. A number of oral magnesium salts are available that vary in their content of elemental magnesium and their oral bioavailability, but little is known about their relative efficacy. Importantly, all of them cause diarrhea, which limits the dose that can be used. Magnesium hydroxide and magnesium oxide are alkalinizing salts with the potential to cause systemic alkalosis, whereas the sulfate and gluconate salts, which present nonreabsorbable anions to the collecting duct, may potentially exacerbate potassium wasting.

A typical daily dose in a patient with normal kidney function and severe hypomagnesemia is 10 to 40 mmol of elemental magnesium in divided doses. Half of this may be sufficient in cases of mild hypomagnesemia, whereas in patients with intestinal magnesium malabsorption, the dose may need to be increased twofold to fourfold. Sustained-release preparations, such as magnesium chloride (Mag-Delay, Slow-Mag) and magnesium L-lactate sustained release (Mag-Tab SR) have the advantage that they are slowly absorbed and thereby minimize kidney excretion of the administered magnesium. By allowing the use of lower doses, these preparations minimize diarrhea.

Potassium-Sparing Diuretics

In patients with inappropriate kidney magnesium wasting, potassium-sparing diuretics that block the distal tubule epithelial sodium channel, such as amiloride and triamterene, may reduce kidney magnesium losses.³³⁰ These drugs may be particularly useful in patients who are refractory to oral repletion or who require such high doses of oral magnesium that diarrhea develops. In rats, amiloride and triamterene have been demonstrated to reduce kidney magnesium clearance at baseline and after induction of magnesium diuresis by furosemide, but the mechanism is unknown. One possibility is that these drugs, by reducing luminal sodium uptake and inhibiting the development of a negative luminal transepithelial potential difference, may favor passive reabsorption of magnesium in the late distal tubule or collecting duct.³³¹

HYPERMAGNESEMIA

CAUSES

In states of body magnesium excess, the kidney has a very large capacity for magnesium excretion. Once the apparent kidney threshold is exceeded, most of the excess filtered magnesium is excreted unchanged into the final urine; the serum magnesium concentration is then determined by the GFR. Thus, hypermagnesemia generally occurs only in two clinical settings, compromised kidney function and excessive magnesium intake.

Kidney Disease

In CKD, the remaining nephrons adapt to the decreased filtered load of magnesium by markedly increasing their fractional excretion of magnesium.³³² As a consequence, serum magnesium levels are usually well maintained until the creatinine clearance falls below about 20 mL/min.³³² Even in advanced kidney disease, significant hypermagnesemia is rare unless the patient has received exogenous magnesium in the form of antacids, cathartics, or enemas. Increasing age is an important risk factor for hypermagnesemia in individuals with apparently normal kidney function; it presumably reflects the decline in GFR that normally accompanies older age.³³³

Excessive Magnesium Intake

Hypermagnesemia can occur in individuals with a normal GFR when the rate of magnesium intake exceeds the kidney excretory capacity. It has been reported with excessive oral ingestion of magnesium-containing antacids³³⁴ and cathartics,³³⁵ with the use of rectal magnesium sulfate enemas,³³⁶ and is common with large parenteral doses of magnesium, such as those given for preeclampsia. Toxicity from enterally administered magnesium salts is particularly common in patients with inflammatory disease, obstruction,³³⁴ or perforation of the gastrointestinal tract, presumably because magnesium absorption is enhanced.

Miscellaneous Causes

Modest elevations in serum magnesium levels have occasionally been described in patients receiving lithium therapy, as well as in postoperative patients and in those with bone metastases, milk-alkali syndrome, familial hypocalciuric hypercalcemia,²⁸⁴ hypothyroidism, pituitary dwarfism, and Addison disease. In most cases, however, the mechanism is unknown.

CLINICAL MANIFESTATIONS

Magnesium toxicity is a serious and potentially fatal condition. Progressive hypermagnesemia is usually associated with a predictable sequence of symptoms and signs.³³⁷ Initial manifestations, observed once the serum magnesium level exceeds 4 to 6 mg/dL, are hypotension, nausea, vomiting, facial flushing, urinary retention, and ileus. If untreated, it may progress to flaccid skeletal muscular paralysis and hyporeflexia, bradycardia and bradyarrhythmias, respiratory depression, coma, and cardiac arrest. An abnormally low (or even negative) serum anion gap may be a clue to hypermagnesemia,³³³ but it is not consistently observed and probably depends on the nature of the anion that accompanies the excess body magnesium.

Cardiovascular System

Hypotension is one of the earliest manifestations of hypermagnesemia,³³⁸ is often accompanied by cutaneous flushing, and is thought to be due to the vasodilation of vascular smooth muscle and inhibition of norepinephrine release by sympathetic postganglionic nerves. Electrocardiographic changes are common but nonspecific.³³⁸ Sinus or junctional bradycardia may develop, as well as varying degrees of sinoatrial, atrioventricular, and His bundle conduction block. Cardiac arrest as a result of asystole is often the terminal event.

Nervous System

High levels of extracellular magnesium inhibit acetylcholine release from the neuromuscular end plate,³³⁹ leading to the development of flaccid skeletal muscle paralysis and hyporeflexia when the serum magnesium level exceeds 8 to 12 mg/dL. Respiratory depression is a serious complication of advanced magnesium toxicity.³³⁸ Smooth muscle paralysis also occurs and is manifested as urinary retention, intestinal ileus, and pupillary dilation. Signs of central nervous system depression, including lethargy, drowsiness, and eventually coma, have been well described in cases of severe hypermagnesemia, but may also be entirely absent.

Treatment

Mild cases of magnesium toxicity in individuals with good kidney function may require no treatment other than cessation of magnesium supplements because kidney magnesium clearance is usually quite rapid. The normal half-life of serum magnesium is approximately 28 hours. In the event of serious toxicity, particularly cardiac toxicity, temporary antagonism of the effect of magnesium may be achieved by the administration of IV calcium (1 g of calcium chloride infused into a central vein over a period of 2 to 5 minutes, or calcium gluconate infused through a peripheral vein, repeated after 5 minutes if necessary).³³⁷ Kidney excretion of magnesium can be enhanced by saline diuresis and the administration of furosemide, which inhibits tubular reabsorption of magnesium in the medullary TAL.

In patients with ESKD, the only way to clear the excess magnesium may be by dialysis or hemofiltration. The typical dialysate for hemodialysis contains 0.6 to 1.2 mg/dL of magnesium, but magnesium-free dialysate can also be used and is generally well tolerated, except for muscle cramps.³⁴⁰ Hemodialysis is extremely effective at removing excess

magnesium and can achieve clearances of up to 100 mL/min.³⁴⁰ As a rough rule of thumb, the expected change in the serum magnesium level after a 3- to 4-hour dialysis session with a high-efficiency membrane is approximately one-third to one-half the difference between the dialysate Mg^{2+} concentration and predialysis serum ultrafilterable magnesium (estimated at 70% of total serum magnesium).³⁴⁰ Note that when hemodialysis is performed using a bath with the same total concentration of magnesium as in serum, the net transfer of magnesium into the patient occurs because the ultrafilterable (and therefore free) magnesium concentration in serum is less than the total concentration; thus, the gradient of free Mg^{2+} is directed from the dialysate to blood.

DISORDERS OF PHOSPHATE HOMEOSTASIS

Body phosphate metabolism is regulated through plasma inorganic phosphorus (Pi) concentration. Of the total body phosphorus content (500–800 g), 85% is in the skeleton, 14% in soft tissues, and the rest is distributed between other tissues and ECF. Of the Pi contained in bone, roughly 200 mg is recycled daily.¹ Two-thirds of the phosphorus in blood exists as organic phosphates (mainly phospholipids) and one-third as Pi. Pi in blood, for practical purposes, involves two orthophosphates, $H_2PO_4^-$ and HPO_4^{2-} . At a plasma pH of 7.4, there are four divalent HPO_4^{2-} ions for every one monovalent $H_2PO_4^-$ ion, so the composite valence is 1.8 (i.e., 1 mmol Pi = 1.8 mEq). Thus, Pi in plasma circulates as phosphates, but is measured in the laboratory as phosphorus (normal values, 2.5–4.5 mg/dL). There are great variations in the normal range of plasma Pi levels with age, from up to 7.4 mg/dL in infants and up to 5.8 mg/dL in children between 1 and 2 years of age.^{4,341} Even in adults, there is a gradual decline in plasma Pi with age, although postmenopausal women, in general, tend to have slightly higher plasma Pi levels compared with their male counterparts.³⁴² Between 85% and 90% of plasma phosphorus is filterable by the kidneys (50% as ionized Pi and 40% complexed with cations), and the remainder is bound to plasma proteins.

The average daily phosphorus intake varies from 800 to 1500 mg, mostly as Pi. Approximately 60% of this is absorbed by the intestine through active transport and paracellular diffusion. The active transport of intestinal phosphate is primarily through type IIb sodium-phosphate cotransporters (NaPi-IIb) and, to a lesser extent, type III transporters (Pit1 and Pit2).³⁴³ Systemic 1,25(OH)₂D levels and dietary phosphorus are important physiologic regulators of intestinal phosphate absorption, with high 1,25(OH)₂D and low dietary phosphate levels promoting the intestinal uptake of phosphate. However, the discovery of several novel phosphatonins, combined with other studies suggesting the existence of a poorly defined intestine-kidney signaling axis,³⁴⁴ has implied that this process may be more complex than originally appreciated. In addition to its absorption from the small intestine, approximately 150 to 200 mg of phosphorus is secreted daily by the colon.¹

As discussed in detail in Chapter 7, the kidney is the major organ regulating phosphate homeostasis. The net kidney excretion of Pi under steady-state conditions is the same as Pi absorbed by the gastrointestinal tract. Up to 80% of kidney

reabsorption of phosphate occurs in the proximal tubule by means of the NaPi cotransporter family of proteins, type IIa (NaPi-IIa) and type IIc (NaPi-IIc) in the luminal brush border membrane.³⁴⁵ The rest of the urinary phosphate is reabsorbed in the distal tubules or excreted in the urine. PTH increases Pi excretion by decreasing the abundance of NaPi-IIa and NaPi-IIc in the brush border membrane. FGF-23 possesses similar actions as those of PTH to limit phosphate reabsorption in the proximal tubule; however, unlike PTH, FGF-23 blocks kidney 1,25(OH)₂D production by suppressing 1 α -hydroxylase activity and stimulating 24-hydroxylation. A low serum phosphate level stimulates kidney NaPi cotransporters and hence phosphate reabsorption.³⁴⁵ Urinary phosphate excretion can be quantified directly from a 24-hour urine collection or can be estimated by calculating the fractional excretion of filtered phosphate (FE_{Pi}) or the kidney tubular maximum reabsorption of phosphate (TmP) to the GFR ratio³⁴⁶ (in mg/dL):

$$\text{TmP/GFCr} = \text{serum Pi} - (\text{urine Pi} \times [\text{serum Cr}/\text{urine Cr}])$$

The latter method is preferred because the TmP/GFR ratio is independent of kidney function. The normal range for TmP/GFR is 2.6 to 4.4 mg/dL³⁴⁷; lower values indicate a decreased maximum kidney phosphate reabsorption threshold and hence excessive urinary phosphate loss. Of note, similar to plasma phosphate levels, TmP/GFR appears to steadily decline with age but, again, slightly increases in women around the time of menopause.³⁴²

HYPERPHOSPHATEMIA

Hyperphosphatemia is generally defined as a serum phosphate level elevated above 5 mg/dL. For children, the upper range of normal is 6 mg/dL. In infants, phosphorus levels as high as 7.4 mg/dL are considered normal.³⁴¹ The serum phosphorus level usually exhibits diurnal variation, with the lowest levels typically being observed in the later morning and peak levels occurring in the first morning hours.³⁴⁸ In CKD, the lowest concentration occurs at 0800 and highest at 1600 and 0400 hours. The circadian pattern in CKD is modifiable by phosphate intake.

The clinical causes of hyperphosphatemia³⁴⁹ can be broadly classified into one of four groups—reduced kidney excretion of phosphate, exogenous phosphate load, acute extracellular shift of phosphorus, or pseudohyperphosphatemia (Box 18.5).

CAUSES

Decreased Kidney Phosphate Excretion

Reduced Glomerular Filtration Rate. Both AKI and CKD can lead to hyperphosphatemia. In the early stages of kidney injury, elevations in PTH and FGF-23 levels increase the urinary fractional excretion of phosphate to compensate for the declining GFR, thus maintaining plasma Pi levels in the normal range. With further decrements in GFR (as in severe AKI or CKD, stage 4 or 5), the reduced functional nephron mass is insufficient to maintain maximal Pi excretion, resulting in hyperphosphatemia. A detailed discussion of hyperphosphatemia caused by decreased kidney function is not provided here; this topic is extensively reviewed in Chapter 55 in the context of chronic kidney disease—mineral bone disorder.

Box 18.5 Causes of Hyperphosphatemia

Decreased Kidney Excretion of Phosphorus

Chronic kidney disease stages 3–5
 Acute kidney injury
 Hypoparathyroidism, pseudohypoparathyroidism
 Acromegaly
 Tumoral calcinosis

- Fibroblast growth factor 23 (FGF-23) inactivating gene mutation
- *GALNT3* mutation with aberrant FGF-23 glycosylation
- *KLOTHO* inactivating mutation with FGF-23 resistance
- Bisphosphonates

Exogenous Phosphorus Administration

Ingestion of phosphate, phosphate-containing enemas
 Intravenous phosphate delivery

Redistribution of Phosphorus (Intracellular to Extracellular Shift)

Respiratory acidosis, metabolic acidosis
 Tumor lysis syndrome
 Rhabdomyolysis
 Hemolytic anemia
 Catabolic state

Pseudohyperphosphatemia

Hyperglobulinemia
 Hyperlipidemia
 Hyperbilirubinemia
 Medications

- Liposomal amphotericin B
- Rec tissue plasma activator
- Heparin

Hemolysis of blood specimen

Hypoparathyroidism and Pseudohypoparathyroidism. Deficient secretion of PTH (hypoparathyroidism) or kidney resistance to PTH (pseudohypoparathyroidism) decrease the kidney excretion of phosphate, leading to hyperphosphatemia. In these entities, circulating phosphorus generally reaches a higher than normal steady-state level (6–7 mg/dL) and is accompanied by hypocalcemia due to decreased bone resorption and urine calcium losses. Hypoparathyroidism and pseudohypoparathyroidism have been discussed extensively (see earlier).

Acromegaly. Some patients with acromegaly demonstrate hyperphosphatemia. Parathyroid function is usually normal or slightly increased in acromegaly.⁹¹ The hyperphosphatemia observed appears to result from increased proximal tubule phosphate reabsorption. Growth hormone and insulin-like growth factor-1 directly stimulate proximal tubule phosphorus reabsorption and increase the TmP/GFR.³⁵⁰

Familial Tumoral Calcinosis. Familial tumoral calcinosis (FTC; OMIM 211900) is a rare autosomal recessive disorder characterized by hyperphosphatemia and the progressive deposition of calcium phosphate crystals in periarticular and soft tissues. The hyperphosphatemia in this disease results from increased proximal tubular reabsorption of

phosphate, usually due to loss-of-function mutations in the *N*-acetyl- α -D-galactosamine gene (*GALNT3*),³⁵¹ which encodes a glycosyltransferase that is thought to prevent the degradation of intact FGF-23.^{352,353} Additional gene mutations have been linked to tumoral calcinosis, including deactivating mutations in the FGF-23 gene,^{354,355} and in the *KLOTHO* gene, which encodes a cofactor necessary for FGF-23 binding to its receptor.³⁵⁶ FTC has been described mainly in families of African and Mediterranean descent.

The unifying pathogenic mechanism in FTC is abrogation of the phosphaturic effect of FGF-23. As such, missense mutations in FGF-23 inhibit its secretion, mutations in *GALNT3* cause aberrant glycosylation and premature degradation of FGF-23,³⁵² and mutations in *KLOTHO* lead to end-organ resistance to FGF-23. Similar to humans, mice lacking FGF-23 or *KLOTHO* expression exhibit severe hyperphosphatemia associated with extensive soft tissue calcification.³⁵⁷ Thus, FTC may be the phenotypic opposite of X-linked and autosomal dominant hypophosphatemic rickets (see later discussion).

PTH, alkaline phosphatase, and calcium levels in FTC are commonly in the normal range. On the other hand, serum 1,25(OH)₂D levels are often increased.³⁵⁸ Together, a normal serum calcium concentration and increased serum phosphorus level lead to an elevated calcium-phosphate product and the slow tissue deposition of calcium phosphate crystals. Decreasing the intestinal absorption of phosphate, either by decreasing dietary phosphorus intake or adding phosphate binders such as sevelamer, is standard therapy for FTC. Additional studies have indicated that the chronic use of acetazolamide may increase urinary phosphate wasting and reduce calcium-phosphate deposits in these patients.³⁵⁹

Bisphosphonates. Bisphosphonates generally cause mild hyperphosphatemia by altering systemic phosphate distribution and decreasing urinary phosphate excretion.^{360,361} The levels of PTH and response with urinary excretion of cAMP to exogenous PTH are normal in patients receiving bisphosphonates.

Exogenous Phosphate Load

Severe hyperphosphatemia has been recognized for at least a half-century as a complication of sodium phosphate taken orally as a cathartic agent or as a sodium phosphate monobasic or dibasic rectal enema (Fleet enema).³⁴⁹ Acute kidney injury, hypocalcemia, severe electrolyte disturbances, and death have been described in response to high-dose sodium phosphate administration. Despite these reports, sodium phosphate preparations remain widely used for bowel preparation prior to colonoscopy. Although the risk for hyperphosphatemia is most pronounced in patients with underlying CKD, even healthy individuals can experience a substantial rise in serum Pi levels (up to 9.6 mg/dL in one prospective study) following sodium phosphate therapy.³⁶² Over the last 10 years, numerous case reports and case series have described an association between the use of oral sodium phosphate for bowel cleansing and phosphate nephropathy. The typical presentation of phosphate nephropathy involves an acute deterioration in kidney function days to weeks following a colonoscopy. Kidney biopsy shows acute and chronic tubular injury, with calcium phosphate deposits (tubular calcifications).^{363,364} Most patients will not recover kidney function fully, and some progress to end-stage disease. Overall, such

adverse kidney effects are uncommon in patients treated with sodium phosphate but, for those select few who are affected, consequences may be serious. The risk factors for developing phosphate nephropathy are advanced age, female gender, impaired kidney function, volume contraction, ulceration of bowel mucosa, bowel obstruction or ileus, hypertension, and use of angiotensin-converting enzyme inhibitors (ACEIs), angiotensin receptor blockers (ARBs), and nonsteroidal antiinflammatory drugs (NSAIDs).^{363,365} Alternative methods of bowel preparation should be considered for patients with these risk factors.

Less commonly, hyperphosphatemia can be observed in the ICU setting when an excessive amount of IV phosphate is given for hyperalimentation, particularly in patients with ESKD. Similarly, the administration of high-dose fosphenytoin for seizure treatment in the setting of kidney dysfunction has been associated with hyperphosphatemia because phosphate is one of the major metabolites of this drug.³⁶⁶ Finally, vitamin D intoxication can result in hyperphosphatemia, largely because of the simultaneous suppression of PTH production and stimulation of intestinal absorption of phosphate.

Intracellular to Extracellular Shift of Phosphorus

Respiratory Acidosis and Metabolic Acidosis. Respiratory acidosis can lead to hyperphosphatemia, kidney resistance to the effects of PTH, and hypocalcemia.³⁶⁷ The effect is more pronounced in acute respiratory acidosis than in the chronic form. Respiratory acidosis does not appear to alter the kidney handling of phosphorus significantly. Rather, efflux of phosphate from cells into the extracellular space is probably responsible for the hyperphosphatemia of respiratory acidosis.³⁶⁸

Lactic acidosis and, to lesser extent, diabetic ketoacidosis also cause hyperphosphatemia.^{369,370} Metabolic acidosis in general reduces glycolysis and Pi utilization. In lactic acidosis, this effect is intensified by tissue hypoxia and intracellular Pi release. Patients with uncontrolled diabetes mellitus are intracellularly phosphate-depleted, despite hyperphosphatemia, an abnormality that becomes unmasked once insulin therapy is initiated.

Tumor Lysis and Rhabdomyolysis. Because phosphate is predominantly stored intracellularly, clinical conditions associated with increased catabolism and tissue destruction, such as rhabdomyolysis, fulminant hepatitis, hemolytic anemia, severe hyperthermia, and tumor lysis syndrome, often result in hyperphosphatemia. The severity of the hyperphosphatemia may be exacerbated by the development of AKI.

Tumor lysis syndrome is a constellation of metabolic abnormalities such as hyperuricemia, hyperkalemia, and hyperphosphatemia caused by rapid and massive breakdown of tumor cells.^{371,372} Clinical consequences may include AKI, pulmonary edema, cardiac arrhythmia, and seizures. The syndrome typically occurs from 3 days before to 7 days after the initiation of chemotherapy. Hyperphosphatemia is an extremely common clinical finding following treatment for Burkitt lymphoma, especially in patients with preexisting kidney disease; it is also seen in other forms of lymphoma, lymphoblastic and myelogenous leukemias and in patients with solid cancers characterized by a high tumor burden. Malignant lymphoid cells may contain up to four times more

intracellular phosphorus compared with mature lymphoid cells, which explains the high prevalence of hyperphosphatemia following chemotherapy in patients with lymphoid malignancies. The lactate dehydrogenase level before the initiation of therapy appears to correlate with the development of hyperphosphatemia and azotemia in these patients.³⁷³ Phosphate nephropathy with tubular calcifications has been reported in tumor lysis syndrome cases exhibiting extremely high serum Pi levels.³⁷¹

To prevent tumor lysis syndrome, intensive volume expansion is generally recommended before chemotherapy to induce a high urine output (120–150 mL/hour) and phosphate and uric acid excretion.³⁷⁴ The usefulness of urine alkalization (pH >7) with bicarbonate infusion and/or acetazolamide is unclear, and its practice is controversial. Alkalization may increase uric acid solubility in the tubules but requires caution; nephrocalcinosis can occur with aggressive alkalization of the urine because calcium phosphate crystals usually precipitate in alkaline urine. Phosphate binders can be used to decrease the intestinal absorption of phosphate in patients who maintain their oral intake during chemotherapy, but the utility of these drugs is limited in this setting. In severe cases of tumor lysis syndrome, hemodialysis may be necessary to control severe hyperphosphatemia and associated metabolic derangements, with CRRT being most efficacious for maintaining phosphate homeostasis.

Pseudohyperphosphatemia

Spurious measurements of high plasma phosphorus levels may occur under certain conditions as a result of interference with the analytic method used. This problem is most common in the case of paraproteinemia (as occurs in multiple myeloma or Waldenström macroglobulinemia).³⁷⁵ This phenomenon can also be observed in patients being treated with liposomal amphotericin B,³⁷⁶ recombinant tissue plasminogen activator,³⁷⁷ or heparin therapy.³⁷⁸ Spurious phosphate readings can also occur in hemolyzed specimens or in samples from patients with severe hyperlipidemia or hyperbilirubinemia.

CLINICAL MANIFESTATIONS AND TREATMENT

Most of the acute clinical manifestations of hyperphosphatemia stem from hypocalcemia, discussed earlier in this chapter. Chronic hyperphosphatemia of CKD and its metabolic consequences are discussed in Chapter 55. Little is known about the effects of chronic hyperphosphatemia in the absence of CKD because this is very rarely observed.

Treatment of chronic hyperphosphatemia is generally accomplished through dietary phosphate restriction, oral phosphate binders, and kidney replacement therapy. Acute hyperphosphatemia in association with hypocalcemia requires rapid attention. Discontinuation of supplemental phosphates and initiation of hydration are indicated for patients with acute exogenous Pi overload and intact kidney function. Volume expansion can significantly increase urinary phosphate excretion, but plasma calcium levels must be followed closely because further hypocalcemia may occur due to hemodilution. Acetazolamide administration may also increase urinary phosphate excretion.^{379–381} Severe hyperphosphatemia in patients with reduced kidney function or AKI, particularly in those with tumor lysis syndrome, may require kidney replacement therapy. In patients with respiratory or metabolic acidosis, treatment of the underlying acidosis corrects the

phosphate derangement. Similarly, in diabetic ketoacidosis, treatment with insulin and correction of metabolic acidosis rapidly reverses the hyperphosphatemia.

HYPOPHOSPHATEMIA

Hypophosphatemia is a decrease in the concentration of Pi in plasma, and phosphate depletion is a decrease in the total body content of phosphorus. Hypophosphatemia can occur in the presence of a low, normal, or high total body phosphorus content. Similarly, total body phosphate depletion may exist with low, normal, or high plasma Pi levels. The incidence of hypophosphatemia is from 0.2% to 2.2% in hospitalized populations but may be present in up to 30% of chronic alcoholics, 28% to 34% of ICU patients, and 65% to 80% of patients with sepsis.^{382,383} There is an association between hypophosphatemia and in-hospital mortality among hospitalized patients³⁸⁴ and all-cause mortality in the dialysis population.³⁸⁵ It is unclear if hypophosphatemia is a direct contributor to these outcomes.

CLINICAL MANIFESTATIONS

Moderate hypophosphatemia (plasma Pi, 1.0–2.5 mg/dL) usually occurs without significant phosphate depletion and without specific signs or symptoms. In severe hypophosphatemia (plasma Pi < 1.0 mg/dL), phosphate depletion is typically present and can have significant clinical consequences.

The clinical manifestations of hypophosphatemia and phosphate depletion generally result from a decrease in intracellular ATP levels. In addition, erythrocytes experience a decrease in 2,3-diphosphoglycerate levels, which increases hemoglobin-oxygen affinity and prevents efficient oxygen delivery to tissues.³⁸⁶ Hypophosphatemia causes a rise in intracellular cytosolic calcium in cells such as leukocytes, pancreatic islets, and synaptosomes isolated from phosphate-depleted animals. These elevated cytosolic calcium levels have been associated with decreased ATP levels and impaired cell response to stimuli.³⁸⁷

Hematologic consequences include a predisposition to hemolysis, thought to result from increased red cell rigidity.^{388,389} Spontaneous hemolysis is rarely observed with hypophosphatemia alone, but hypophosphatemia can predispose to red cell lysis in the presence of other risk factors. Phagocytosis and chemotaxis of polymorphonuclear cells are diminished because impaired ATP production decreases the phagocytic capability of these cells.³⁹⁰

Severe hypophosphatemia can result in numerous neuromuscular and skeletal abnormalities, including proximal myopathy, bone pain,^{382,391} and rhabdomyolysis.³⁹² Because cell breakdown may lead to the release of intracellular phosphate, normophosphatemia or hyperphosphatemia in this setting may mask the existence of true phosphate depletion. Overt heart failure and respiratory failure as a result of decreased muscle performance may also be observed.^{393,394} Correction of the Pi level leads to improvement of myocardial and pulmonary function.^{395,396} Neurologic manifestations of severe hypophosphatemia include paresthesia, tremor, and encephalopathy; these also improve with phosphate replacement.³⁸² Chronic phosphate depletion alters bone metabolism, leading to increased bone resorption and severe mineralization defects that impair bone structure and strength (e.g., osteomalacia, rickets).

Chronic hypophosphatemia can also lead to proximal and distal kidney tubule defects resulting in water diuresis, glucosuria, bicarbonaturia, hypercalciuria, and hypermagnesuria.³⁹⁷ The hypercalciuria is not solely the result of altered kidney calcium handling but also reflects increased calcium release from bone, consequent to phosphate mobilization, and increased intestinal calcium absorption in response to the accompanying elevation in plasma $1,25(\text{OH})_2\text{D}$ levels.³⁹⁸ Metabolic consequences of hypophosphatemia include insulin resistance, diminished gluconeogenesis, hypoparathyroidism, and metabolic acidosis, with reduced H^+ excretion and ammonia generation. Hypophosphatemia is also a potent stimulator of 1α -hydroxylation to convert $25(\text{OH})\text{D}$ to $1,25(\text{OH})_2\text{D}$.

DIAGNOSIS

The cause of hypophosphatemia can often be delineated from the clinical history or physical examination alone. Shifts of phosphorus from the extracellular to intracellular space usually occur in the setting of an acute illness or treatment (e.g., respiratory alkalosis, treatment of diabetic ketoacidosis). Thus, hypophosphatemia in hospitalized patients typically results from shifts of phosphorus into the intracellular compartment as opposed to kidney losses of phosphorus.³⁹⁹

In situations in which the underlying diagnosis is not immediately apparent, it can be clinically useful to determine the rate of urine phosphate excretion by quantification in a 24-hour urine collection or by calculation of the FE_{Pi} or TmP/GFR . A 24-hour urine phosphate greater than 100 mg, FE_{Pi} greater than 5%, or TmP/GFR less than 2.5 mg/100 mL indicates inappropriate urinary phosphate wasting.³⁴⁷ A high urine phosphate level in the presence of hypophosphatemia results from an acquired defect (e.g., PHPT) or genetic defect (e.g., X-linked hypophosphatemic rickets) in phosphate reabsorption by the proximal tubule.

CAUSES

Hypophosphatemia may be due to increased kidney phosphate excretion, decreased intestinal absorption, shift of phosphate from extracellular to intracellular fluid, or a combination of these mechanisms (Box 18.6).

Increased Kidney Phosphate Excretion

Hypophosphatemia caused by increased urinary phosphate excretion is generally the result of excess PTH, increased production or activity of FGF-23 from normal or dysplastic bone, or a disorder of kidney phosphate handling in the proximal tubule. The recent discovery of several inherited disorders of renal phosphate wasting has contributed to the elucidation of the underlying mechanisms (Fig. 18.5).

Hyperparathyroidism. Both primary and secondary HPT may lead to hyperphosphaturia and hypophosphatemia. PHPT has been discussed earlier in this chapter. The degree of hypophosphatemia observed is usually mild to moderate in severity; increased urinary phosphate excretion is balanced by the mobilization of Pi from the bone and enhanced intestinal absorption of Pi. Secondary HPT resulting from vitamin D deficiency causes hypophosphatemia, not only through the promotion of urinary phosphate wasting by PTH, but also through decreased intestinal absorption of phosphate from low vitamin D levels. The secondary HPT observed in

Box 18.6 Causes of Hypophosphatemia

Increased Urinary Phosphate Excretion

Increased production or activity of fibroblast growth factor 23 (FGF-23)

- Inherited disorders
 - X-linked hypophosphatemia (*PHEX* mutations)
 - Autosomal dominant hypophosphatemic rickets (FGF-23 mutations)
 - Autosomal recessive hypophosphatemic rickets (*DMP1* and *ENPP1* mutations)
- Acquired disorder
 - Tumor-induced osteomalacia

Disorders of proximal tubule Pi reabsorption

- Hereditary hypophosphatemic rickets with hypercalciuria (*SLC34A3* mutations)
- Autosomal recessive kidney phosphate wasting (*SLC34A1* mutations)
- *NHERF1* mutations
- *KLOTHO* mutations
- Fanconi syndrome
- Primary and secondary hyperparathyroidism
- Postkidney transplantation
- Medications—acetazolamide, calcitonin, glucocorticoids, diuretics, bicarbonate, acetaminophen, iron (intravenous), antineoplastics, antiretrovirals, aminoglycosides, anticonvulsants
- Acute tubular necrosis recovery, posturinary obstruction
- Miscellaneous—posthepatectomy, colorectal surgery, volume expansion, osmotic diuresis

Decreased Intestinal Absorption of Phosphate

Malnutrition with low phosphate intake, anorexia, starvation
Malabsorption of phosphate—chronic diarrhea, gastrointestinal tract diseases

Intake of phosphate-binding agents

Vitamin D deficiency or vitamin D resistance

- Nutritional deficiency—low dietary intake, low sun exposure
- Malabsorption
- Chronic kidney disease
- Chronic liver disease
- Vitamin D synthesis and vitamin D receptor defects

Altered Phosphorus Distribution, Intracellular Shift

Acute respiratory alkalosis

Refeeding of malnourished patients, alcoholics

Hungry bone syndrome (postparathyroidectomy)

Hypophosphatemia Resulting From Multiple Mechanisms

Alcoholism

Diabetic ketoacidosis, insulin therapy

Miscellaneous

- Tumor consumption of phosphate—leukemia blast crisis, lymphoma
- Sepsis
- Heat stroke and hyperthermia

patients with CKD is typically associated with hyperphosphatemia because of a decreased ability of the kidney to excrete phosphorus.

Increased Production or Activity of Phosphatonins. There are several rare syndromes of kidney phosphate wasting

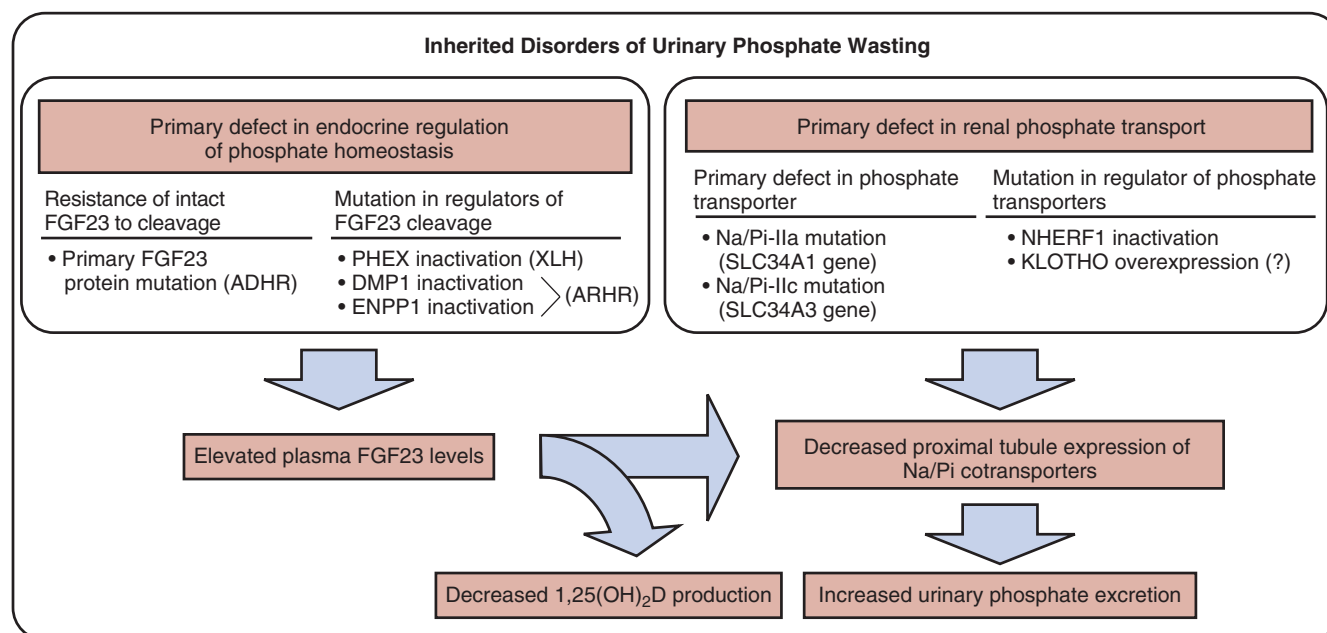


Fig. 18.5 Summary of the inherited disorders of urinary phosphate wasting. Inherited diseases characterized by phosphaturia and hypophosphatemia can occur from a defect in endocrine pathways involved in the systemic regulation of phosphate homeostasis or from a direct mutation in local regulators of renal phosphate transport. *ADHR*, Autosomal dominant hypophosphatemic rickets; *ARHR*, autosomal recessive hypophosphatemic rickets; *DMP1*, dentin matrix protein 1; *ENPP1*, ectonucleotide pyrophosphatase/phosphodiesterase 1; *FGF23*, fibroblast growth factor 23; *Na/Pi-IIa*, type IIa sodium-phosphate cotransporter; *Na/Pi-IIc*, type IIc sodium-phosphate cotransporter; *NHERF1*, sodium-hydrogen exchanger regulatory factor; *PHEX*, phosphate-regulating gene with homology to endopeptidases on the X chromosome; *XLH*, X-linked hypophosphatemic rickets.

associated with rickets or osteomalacia that result from increased production or activity of FGF-23 or other phosphatonins (see Fig. 18.4).⁴⁰⁰

X-Linked Hypophosphatemia. X-linked hypophosphatemia (XLH; OMIM 307800) is a rare X-linked dominant disorder characterized by hypophosphatemia, rickets and osteomalacia, growth retardation, decreased intestinal calcium and phosphate absorption, and decreased kidney phosphate reabsorption. Because it is an X-linked dominant disorder, only one copy of the defective gene is sufficient to cause the disorder when inherited from a parent who has the disorder. Serum 1,25(OH)₂D levels are inappropriately normal or low, and calcium and PTH levels are normal in patients with XLH. It differs from most cases of rickets in that vitamin D supplementation does not cure it. The prevalence of the disease is 1:20,000, penetrance is high, and both females and males are affected.⁴⁰¹

The gene responsible for this disorder, a phosphate-regulating gene with homology to endopeptidases on the X chromosome (*PHEX*), was identified by positional cloning.⁴⁰² Binding of *PHEX* to another bone-derived protein, dentin matrix protein 1 (DMP1), appears to be critical for the suppression of osteocyte production of FGF-23 in bone.⁴⁰³ Inactivating mutations in *PHEX* or DMP1 result in higher circulating levels of FGF-23 and resultant phosphate wasting.^{402,404,405} Abnormal synthesis of 1,25(OH)₂D in XLH can be explained by increased levels of FGF-23, which suppress kidney 1 α -hydroxylase activity.⁴⁰⁶

Treatment of XLH patients with oral phosphate and calcitriol improves their growth rate; however, these therapies

do not reduce kidney phosphate excretion. As a result, the major goal of therapy in these patients has been to allow normal growth and reduce bone pain.⁴⁰⁷ In 2018, the FDA approved burosumab, a human anti-FGF23 monoclonal antibody that appears to be effective for treatment of XLH in children 1 year and older. Burosumab offers a more effective therapeutic strategy, with limited side effects. Studies of burosumab for tumor-induced osteomalacia are in progress.⁴⁰⁸ Thus, FGF23-neutralizing antibodies hold considerable promise as a future treatment for XLH patients.

Autosomal Dominant Hypophosphatemic Rickets. Autosomal dominant hypophosphatemic rickets (ADHR, OMIM 193100) is an extremely rare disorder of phosphate wasting, with a clinical phenotype similar to that of XLH. Some individuals initially present in childhood with hypophosphatemia associated with lower extremity deformities, whereas others present in adolescence or adulthood with bone pain, weakness, and phosphate wasting. In some individuals with early-onset disease, the phosphate wasting returns to normal after puberty.

The ADHR locus has been mapped to chromosome 12p13.3 and identified as FGF-23. Missense mutations of FGF-23 appear to interfere with its proteolytic cleavage by furin or other subtilisin-like proprotein convertases,^{409–411} causing prolonged or enhanced FGF-23 action on the kidney. Treatment is with phosphate replacement and calcitriol, similar to that for patients with XLH.

Autosomal Recessive Hypophosphatemic Rickets. Reports describing families with autosomal recessive forms of hypophosphatemic rickets have also emerged. As noted,

inactivating mutations in the gene encoding DMP1 lead to a hypophosphatemic syndrome in humans.⁴¹² Similarly, a more recently described mutation in ectonucleotide pyrophosphatase/phosphodiesterase 1 (ENPP1), a protein responsible for the conversion of extracellular ATP into inorganic pyrophosphate,⁴¹³ has also been demonstrated to result in a phosphate-wasting syndrome in humans.⁴¹⁴ Interestingly, the deletion of ENPP1 in mice not only leads to increased FGF-23 production by bone and associated urinary phosphate wasting, but also to defective bone mineralization and soft tissue calcification.⁴¹⁵ To date, no studies have elucidated how ENPP1 or pyrophosphate may regulate FGF-23 production.

Tumor-Induced Osteomalacia. Tumor-induced osteomalacia (TIO), or oncogenic osteomalacia, is an acquired paraneoplastic syndrome of kidney phosphate wasting. Usually this syndrome presents in older adults, with a protracted course. Hypophosphatemia with normal serum calcium and PTH levels, kidney Pi wasting, low calcitriol levels, and decreased bone mineralization are the hallmarks of TIO.⁴¹⁶ It is caused by mesenchymal tumors that express and secrete FGF-23.⁴¹⁷ In addition to FGF-23, other phosphaturic factors are often secreted by these tumors, including matrix extracellular phosphoglycoprotein (MEPE), frizzled-related protein 4 (FRP-4), and FGF-7.⁴¹⁸

The definitive treatment of TIO is complete resection of the inciting tumor; however, these mesenchymal tumors are often small and difficult to localize. Medical treatment with phosphate supplementation and calcitriol is frequently necessary to improve bone healing in patients for whom tumor localization or resection is unsuccessful. Cinacalcet has also been used to induce hypoparathyroidism and decrease phosphate wasting, with good response,⁴¹⁹ although hypocalcemia is always a concern when using this therapy in patients with normal kidney function.

Disorders of Proximal Tubule Inorganic Phosphorus Reabsorption

Hereditary Hypophosphatemic Rickets With Hypercalciuria. Hereditary hypophosphatemic rickets with hypercalciuria (HHRH; OMIM 241530) is a rare autosomal recessive syndrome characterized by rickets, short stature, kidney phosphate wasting, and hypercalciuria. HHRH is caused by mutations in *SLC34A3*, the gene encoding the kidney sodium-phosphate cotransporter NaPi-2c.⁴²⁰ Patients have an appropriate elevation in 1,25(OH)₂D levels, which results in hypercalciuria. They are treated with phosphorus supplementation.

A similar autosomal recessive disorder has been described in two patients with loss-of-function mutations in the *SLC34A1* gene, which encodes the NaPi-IIa sodium phosphate cotransporter. However, unlike HHRH, these patients also had Fanconi syndrome (see later).⁴²¹

More recently, gene mutations in the sodium-hydrogen exchanger regulatory factor 1 (NHERF1) have been described in patients with hypophosphatemia and nephrolithiasis from kidney phosphate wasting.⁴²² NHERF1 is a protein that plays an essential role in the delivery of sodium phosphate cotransporters to the apical membrane in the proximal tubule.⁴²³ Similarly, a de novo translocation mutation in the *KLOTHO* gene has also been linked to a syndrome characterized by elevated plasma α -klotho levels, hypophosphatemia, and

HPT.⁴²⁴ The membrane-bound form of klotho is expressed locally in the kidney in the proximal and distal tubules; prior studies have suggested that klotho has an independent role in kidney phosphate transport.⁴²⁵ However, the exact mechanisms responsible for the clinical phenotype in this patient exhibiting a translocation mutation of the *KLOTHO* gene remain undetermined.

Fanconi Syndrome. Fanconi syndrome is a disorder characterized by generalized proximal tubule dysfunction leading to defects in the reabsorption of glucose, phosphate, calcium, amino acids, bicarbonate, uric acid, and other organic compounds.⁴²⁶ This syndrome can be genetic or acquired. Inherited causes of Fanconi syndrome include cystinosis, tyrosinemia, and Wilson disease. Acquired causes include monoclonal gammopathies, amyloidosis, collagen vascular diseases, kidney transplant rejection, and many drugs or toxins, such as heavy metals, antineoplastic agents, antiretroviral agents, aminoglycosides, and anticonvulsants.^{427,428} Over time, severe hypophosphatemia in Fanconi syndrome can lead to defective bone mineralization, with increased fracture risk. Of interest, a recessive loss-of-function mutation in the *SLC34A1* gene encoding the NaPi-IIa proximal tubule transporter was shown to cause Fanconi syndrome, including phosphaturia and hypophosphatemia and other perturbations in proximal tubule transport, perhaps arising from the accumulation of the misfolded gene product and consequent endoplasmic reticulum-associated stress response.⁴²⁹

Kidney Transplantation. Hypophosphatemia is observed in up to 90% of patients after kidney transplantation.^{430,431} It is typically mild to moderate and mostly occurs during the first weeks after surgery, but may persist for months to years.⁴³⁰ These patients have phosphaturia and decreased TmP/GFR, with a well-preserved GFR. The causes of posttransplantation hypophosphatemia include persistent (tertiary) HPT,⁴³⁰ excess of FGF-23 in the posttransplantation period,⁴³² 25(OH)D and 1,25(OH)₂D deficiency, and immunosuppressive medication.⁴³³ Pretransplantation levels of PTH and FGF-23 predict the severity of hypophosphatemia, and both hormones may act synergistically to increase phosphaturia in this setting.⁴³⁴ Cinacalcet has been shown to correct urinary phosphate wasting and normalize plasma Pi in posttransplantation patients by decreasing PTH levels without an impact on high levels of FGF-23.⁴³⁵ The major consequence of posttransplantation hypophosphatemia is progressive bone loss and osteomalacia. Management of posttransplantation hypophosphatemia includes replacement of phosphate, correction of vitamin D deficiency, and treatment of HPT.

Drug-Induced Hypophosphatemia. There is an extensive and growing list of medications that can cause hypophosphatemia and phosphate urinary losses as part of Fanconi syndrome, as discussed previously, or by affecting only NaPi transporters in the kidney. Diuretics, including acetazolamide, loop diuretics, and some thiazides with carbonic anhydrase activity, such as metolazone, can increase phosphaturia. The volume contraction that accompanies the use of diuretics usually stimulates proximal tubular NaPi reabsorption and prevents the development of severe hypophosphatemia. Conversely, volume expansion with saline can cause phosphaturia and hypophosphatemia.⁴³⁶ Corticosteroids decrease intestinal

phosphorus absorption and increase kidney phosphorus excretion and thus may cause mild to moderate hypophosphatemia.⁴³⁷ Hypophosphatemia has been reported in patients treated for malignancies with many of the novel tyrosine kinase inhibitors, including imatinib (50%),⁴³⁸ sorafenib (13%),⁴³⁹ and nilotinib.⁴⁴⁰ The mechanism is thought to be due to inhibition of calcium and Pi resorption from bone, together with secondary HPT, leading to phosphaturia,⁴⁴¹ or may be caused by the development of a partial Fanconi syndrome.⁴⁴² Administration of parenteral iron formulations containing carbohydrate moieties has been associated with hypophosphatemia, phosphate wasting, and inhibition of 1 α -hydroxylation of vitamin D,^{443,444} which was found to be mediated by an increase in circulating levels of intact FGF-23.⁴⁴⁵ Hypophosphatemia has also been reported in acetaminophen toxicity; the mechanism for this association is likely multifactorial, but appears at least partially to result from urinary phosphate wasting.⁴⁴⁶ Finally, the administration of large doses of estrogens in patients with metastatic prostate carcinoma can also produce hypophosphatemia.⁴⁴⁷

Miscellaneous Causes. Significant urinary losses of phosphate may lead to hypophosphatemia during recovery from acute tubular necrosis and from obstructive uropathy. Postoperative hypophosphatemia has been reported after liver resection, colorectal surgery, aortic bypass, and cardiothoracic surgery.^{448–450} Posthepatectomy hypophosphatemia appears to be due to a transient increase in kidney fractional excretion of phosphate rather than to increased metabolic demand by the regenerative liver.⁴⁵¹

Decreased Intestinal Absorption

Malnutrition. Malnutrition with a low phosphate intake is not a common cause of hypophosphatemia. Increased kidney reabsorption of phosphorus can compensate for all but the most severe decreases in oral phosphate intake. However, if phosphate deprivation is prolonged and severe (<100 mg/day), or if it coexists with diarrhea, the continued colonic secretion of phosphate can lead to hypophosphatemia. Hypophosphatemia seen in children with protein malnutrition and kwashiorkor correlates with increased mortality.⁴⁵²

Malabsorption. More common is hypophosphatemia resulting from malabsorption. Most phosphorus absorption occurs in the duodenum and jejunum, and intestinal disorders affecting the small intestine may lead to hypophosphatemia.⁴⁵³ Phosphate-binding cations such as aluminum, calcium, magnesium, and iron form complexes with phosphorus in the gastrointestinal tract, resulting in decreased phosphate absorption. Hypophosphatemia can develop quickly, even in patients given a relatively moderate but sustained dosage of phosphate binders. When combined with poor nutritional intake or extensive dialysis, this therapy may result in so-called overshoot hypophosphatemia. Prolonged use of phosphate-binding antacids can lead to clinically significant osteomalacia.⁴⁵⁴

Vitamin D–Mediated Disorders. Vitamin D is critical for the normal control of phosphorus. Deficiency of vitamin D leads to decreased intestinal absorption of phosphorus and to hypocalcemia, HPT, and a consequent PTH-mediated increase in kidney phosphorus excretion. Syndromes of vitamin D

deficiency or resistance characterized by hypophosphatemia, hypocalcemia, and bone disease are discussed in an earlier section of this chapter that discusses hypocalcemia.

Redistribution of Phosphate. Redistribution of phosphate from the extracellular space into cells is a common cause of hypophosphatemia in hospitalized patients. This shift of phosphate occurs by various mechanisms, including elevated levels of insulin, glucose, and catecholamines, respiratory alkalosis, increased cell proliferation (leukemia blast crisis, lymphoma), and rapid bone mineralization (hungry bone syndrome).

Respiratory Alkalosis. The fall in carbon dioxide during acute respiratory alkalosis causes carbon dioxide diffusion from the intracellular space, increases intracellular pH, and stimulates glycolysis. The consequent increase in the formation of phosphorylated carbohydrates leads to a decrease in extracellular phosphorus levels.⁴⁵⁵ When the alkalosis is prolonged and severe, phosphorus levels can drop below 1 mg/dL.⁴⁵⁶ Mild hypophosphatemia may occur during the increased ventilation after treatment of an asthma attack⁴⁵⁷ and in patients with panic disorders with intermittent hypocapnia. Hypophosphatemia is common in mechanically ventilated patients, particularly if they are also receiving glucose infusions. The urinary phosphate excretion can drop to undetectable levels, indicating maximal urinary Pi reabsorption.

Refeeding Syndrome. In chronically malnourished individuals, rapid refeeding can result in significant hypophosphatemia. The incidence of refeeding-related hypophosphatemia is high in hospitalized patients receiving parenteral nutrition, as high as one in three in one series.^{458,459}

Risk factors for refeeding syndrome include eating disorders, chronic alcoholism, kwashiorkor, cancer, and diabetes mellitus.⁴⁵⁸ Refeeding after even very short periods of starvation can lead to hypophosphatemia.⁴⁵⁹ The mechanism is related to an insulin-induced increase in cellular phosphate uptake and utilization. The maintenance of serum Pi in the normal range is essential in the management of refeeding syndrome. Adequate phosphate (20–30 mmol of Pi/L) in the parenteral nutrition formulation generally prevents this complication. Even higher amounts may be required for patients with diabetes or chronic alcoholism.

Hypophosphatemia Resulting From Multiple Mechanisms

Alcoholism. Hypophosphatemia and phosphate depletion are particularly common and often a severe problem in alcoholic patients with poor intake, vitamin D deficiency, and heavy use of phosphate-binding antacids.⁴⁶⁰ Alcohol-induced proximal tubule dysfunction also contributes to phosphate depletion.²³⁴ Alcoholics frequently develop acute respiratory alkalosis due to alcohol withdrawal, sepsis, or cirrhosis. Phosphorus deficiency is often not manifested as hypophosphatemia at the initial evaluation for medical care. Typically, refeeding, administration of IV glucose, or both in this patient population stimulates shifts of phosphorus into cells and thereby uncovers severe hypophosphatemia. Hypophosphatemic alcoholics are at high risk for the development of rhabdomyolysis.³⁹²

Diabetic Ketoacidosis. In uncontrolled diabetes, phosphate is released from cells and ultimately appears in the urine because of concomitant glycosuria, ketonuria, acidosis, and osmotic diuresis, which all increase urinary phosphate excretion.⁴⁶¹ Although serum phosphate levels may be normal, total body phosphate stores are usually low. During treatment of diabetic ketoacidosis, the development of hypophosphatemia is extremely common.⁴⁶² Administration of insulin stimulates the cellular uptake of phosphorus, and thus the serum phosphate level can fall dramatically with treatment.³⁷⁰ However, routine administration of phosphate in this setting, before the development of hypophosphatemia, is discouraged because it may lead to significant hypocalcemia.⁴⁶³ Phosphate depletion can itself be a cause of insulin resistance, and a decrease in insulin requirements has been observed after phosphate replacement therapy.^{464,465}

Miscellaneous Disorders. Moderate, and at times severe, hypophosphatemia may be observed in acute leukemia in the leukemic phase of lymphomas⁴⁶⁶ and during hematopoietic reconstitution after stem cell transplantation.⁴⁶⁷ Rapid cell growth, with consequent phosphorus utilization, is very likely responsible for the decrease in extracellular phosphorus. Hypophosphatemia has been observed in a woman with toxic shock syndrome⁴⁶⁸ and is commonly observed in sepsis,⁴⁶⁹ but the complicated clinical picture in septic patients makes it difficult to delineate a specific mechanism. Rapid volume expansion diminishes proximal tubule sodium phosphate reabsorption and may lead to transient hypophosphatemia.⁴³⁶ Hypophosphatemia is seen in patients with heat stroke, as well as hyperthermia, mainly due to increased kidney phosphorus excretion.

TREATMENT

The first step in the management of hypophosphatemia is to establish the cause of the low Pi, followed by a determination of whether Pi replacement is necessary. There is little evidence that mild hypophosphatemia (Pi = 2.0–2.5 mg/dL) has significant clinical consequences in humans or that aggressive Pi replacement is needed. This is particularly true when the Pi shift is the major cause of the hypophosphatemia.

Patients with symptomatic hypophosphatemia and phosphate depletion do require replacement therapy. Those with severe hypophosphatemia, with a plasma Pi level less than 1 mg/dL, even in the absence of phosphate depletion, will need IV phosphate therapy. Because the serum level of phosphorus may not be an accurate reflection of total body stores, it is essentially impossible to predict the amount of phosphorus necessary to correct phosphorus deficiency and hypophosphatemia.⁴⁷⁰ In chronically malnourished patients (e.g., anorectics, alcoholics), significant phosphorus repletion will be necessary, whereas in patients who are hypophosphatemic from other causes (e.g., antacid ingestion, acetazolamide use), correction of the underlying problem may be sufficient.

Phosphate can be administered orally or parenterally. In mild or moderate hypophosphatemia, oral repletion with low-fat milk (containing 0.9 mg Pi/mL) is well tolerated and effective. Alternatively, oral tablets containing 250 mg (8 mmol) of phosphorus from a combination of sodium phosphate and potassium phosphate salts can be prescribed.

A typical patient with moderate to severe hypophosphatemia would probably need 1000 to 2000 mg (32–64 mmol) of phosphorus/day to have body stores repleted within 7 to 10 days. Side effects include diarrhea, hyperkalemia, and volume overload.

IV phosphorus repletion is generally reserved for individuals with severe hypophosphatemia (Pi <1 mg/dL). Various regimens are used in clinical practice, all based on uncontrolled observational studies. Some are more conservative in the amount of phosphate delivered to avoid side effects, which may include kidney failure, hypocalcemic tetany, and hyperphosphatemia. One standard regimen is to administer 2.5 mg/kg body mass of elemental phosphorus (0.08 mmol/kg of phosphate over a 6-hour period for severe asymptomatic hypophosphatemia; 5 mg/kg body mass of elemental phosphorus [0.16 mmol/kg of phosphate] over a 6-hour period for severe symptomatic hypophosphatemia).⁴⁷¹ Others have used a more intensive regimen of 10 mg/kg body mass (0.32 mmol/kg of phosphate) administered over 12 hours.⁴⁷² However, even with these higher doses, only 58% of treated patients achieved serum Pi levels above 2 mg/dL. A graded dosing scheme for IV phosphate replacement (0.16 mmol/kg over 4 to 6 hours, 0.32 mmol/kg over 4 to 6 hours, and 0.64 mmol/kg over 6 to 8 hours for serum Pi levels of 2.3–3.0, 1.6–2.2, and <1.5 mg/dL, respectively) has been used effectively in ICU patients without kidney dysfunction and hypercalcemia.⁴⁷³ Other intensive phosphate replacement regimens have been reported and found to be effective and safe in the ICU for select patients with severe hypophosphatemia.³⁸⁴



Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

1. A 21-year-old man was evaluated for hypocalcemia that was discovered on a routine screening examination in the past year. The patient notes occasional cramping of his hands but no other complaints. His physical examination is normal except for a positive Chvostek sign. His laboratory studies showed the following: serum calcium, 7.5 mg/dL; phosphate, 5.2 mg/dL; and PTH 40 pg/mL. His urine calcium level was 410 mg/24 h. Which ONE of the following choices would be the best for this patient?
 - a. Parathyroidectomy
 - b. Continued observation
 - c. Begin a calcimimetic agent
 - d. Evaluate family members for genetic defects in the CSR

Answer: d

Rationale: Heterozygous gain-of-function mutations in the calcium sensing receptor can activate the calcium sensing receptor or cause calcium sensing receptor to be hyper responsive to extracellular calcium. This condition has been termed autosomal dominant hypoparathyroidism. The phenotype is characterized with mild hypocalcemia, hypomagnesemia and hypercalciuria with low or inappropriately low of PTH. Treatment with calcium supplements and vitamin D is usually warranted for patients with severe symptomatic hypocalcemia.

2. A 39-year-old female patient was noted to have hypercalcemia of 11.9 mg/dL during an evaluation for a recurrent kidney stone episode. Ionized calcium confirms the hypercalcemia. Extensive evaluation including serum and urine protein electrophoresis are negative. iPTH level is 85 pg/mL and the bone density is below normal. At this point, which ONE of the following statements is CORRECT about medical therapy for this patient?
 - a. Patient should be referred to an endocrine surgeon for parathyroidectomy.
 - b. Patient should be started on cinacalcet.
 - c. Patient should be initiated on hormone replacement therapy with estradiol to decrease PTH and calcium levels.
 - d. Patient should be closely observed but no therapeutic intervention should be started.

Answer: a

Rationale: Primary hyperparathyroidism is one of the most common causes of hypercalcemia and the most common cause of hypercalcemia in the outpatient setting. The normal levels of PTH measured by various assays range from 8 to 80 pg/mL. In hypercalcemia of primary hyperparathyroidism, PTH levels are normal or high. Lack of PTH of suppression by hypercalcemia suggest a diagnosis of primary hyperparathyroidism. Surgery is still standard therapy for primary hyperparathyroidism. Parathyroidectomy is indicated for all patients with biochemically confirmed primary hyperparathyroidism who have symptoms or signs of disease, such as

a history of hypercalcemia, chronic kidney disease and/or kidney stones.

3. A 58-year-old man was referred for management of nephrotic syndrome. The laboratory evaluation revealed a GFR 35 mL/min/1.73 m², a urinary protein excretion of 14 g/d, serum albumin of 1.5 g/dL, intact parathyroid hormone level of 200 pg/mL and a 25(OH)D levels < 10 ng/mL. Which one of the following statements is most accurate?
 - a. Vitamin D status is reflected by the serum levels of 1,25(OH)₂D and not 25(OH)D
 - b. 25(OH)D deficiency is highly prevalent in patients with chronic kidney disease
 - c. Low levels of 25(OH)D are not associated with increased parathyroid hormone
 - d. Proteinuria does not contribute to vitamin D deficiency

Answer: b

Rationale: Proteinuric kidney disease is a risk factor for vitamin D deficiency. Current practice guidelines suggest that vitamin D deficiency this should be corrected if PTH levels are elevated.

4. Patient is an 80-year-old-female with stage III chronic kidney disease who is referred to a nephrologist for evaluation and management. Several laboratory tests are ordered. Significant results include an iPTH of 200 pg/mL, serum calcium 8.3 mg/dL, serum phosphate of 4.1 mg/dL and 25(OH)D levels of 12 ng/mL. Which test would intervene first to attempt to normalize iPTH levels?
 - a. Fibroblast growth factor 23
 - b. Klotho
 - c. Serum calcium
 - d. 25(OH)D

Answer: d

Rationale: Abnormalities of vitamin D metabolism play a key role in the development of secondary hyperparathyroidism.

5. A 60-year-old male recently initiated kidney replacement therapy in the form of hemodialysis. Patient's nephrologist noticed that serum phosphate has increased to 6 mg/dL. Last serum calcium is 8.2 mg/dL, and iPTH is within range at 170 pg/mL. Which of the following interventions would be an acceptable course of action?
 - a. Calcium supplementation
 - b. Phosphate binders
 - c. Vitamin D sterols
 - d. Cinacalcet

Answer: b

Rationale: Current guidelines recommend that serum phosphate in patients requiring hemodialysis be maintained in within a normal range using calcium or non-calcium containing binders.